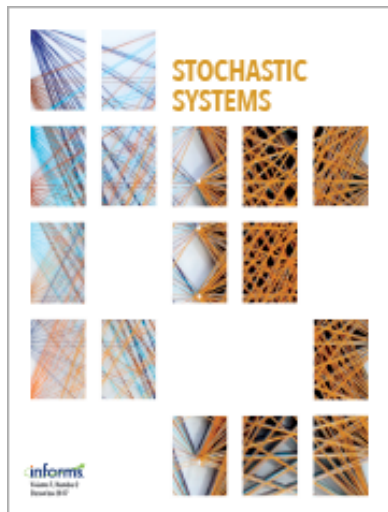




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Splitting Algorithms for Rare Events of Semimartingale Reflecting Brownian Motions

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Abstract. We study rare event simulations of semimartingale reflecting Brownian motions (SRBMs) in an orthant. The rare event of interest is that a d -dimensional positive recurrent SRBM enters the set $\{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq n\}$ before hitting a small neighborhood of the origin $\{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \epsilon\}$ as $n \rightarrow \infty$ with a starting point outside the two sets and of order $o(n)$. We show that, under two regularity conditions (the Dupuis–Williams stability condition of the SRBM and the Lipschitz continuity assumption of the associated Skorokhod problem), the probability of the rare event satisfies a large deviation principle. To study the variational problem (VP) for the rare event in two dimensions, we adapt its exact solution from developed by Avram, Dai, and Hasenbein in 2001. In three and higher dimensions, we construct a novel subsolution to the VP under a further assumption that the reflection matrix of the SRBM is a nonsingular $\mathcal{M}$ -matrix. Based on the solution/subsolution, particle-based simulation algorithms are constructed to estimate the probability of the rare event. Our estimator is asymptotically optimal for the discretized problem in two dimensions and has exponentially superior performance over standard Monte Carlo in three and higher dimensions. In addition, we establish that the growth rate of the relative bias term arising from discretization is subexponential in all dimensions. Therefore, we can estimate the probability of interest with subexponential complexity growth in two dimensions. In three and higher dimensions, the computational complexity of our estimators has a strictly smaller exponential growth rate than the standard Monte Carlo estimators.



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Keywords: semimartingale reflecting Brownian motions • rare event simulations • particle splitting algorithms • large deviation principle • variational problems • subsolutions

1. Introduction

In many scenarios, one is interested in the dynamics of an entity that is constrained to a subset of its possible state space by some mechanism. For example, when studying population levels of many interacting species (e.g., chemical compounds or biological populations), a natural constraint is that all population levels are nonnegative. Alternatively, one may simply be interested in the dynamics of particles in a container from which they cannot escape. The so-called reflecting diffusions are a group of stochastic processes that have been developed to study the phenomena of constrained stochastic dynamics. Roughly speaking, a reflecting diffusion process behaves as an ordinary diffusion process in the interior of its domain, and it is pushed into the domain whenever the boundary of the domain is hit.

A substantial amount of research on reflecting diffusions has been carried out going back to the seminal works of Skorohod (1961) and Kingman (1962) in one dimension. Watanabe (1971) and Stroock and Varadhan (1971) prove the existence of reflecting diffusions in higher dimensions when the domain is sufficiently smooth. When dealing with a particle constrained to an orthant, one does not have a smooth domain, and an important extension is the work of Harrison and Reiman (1981) that studied semimartingale reflecting Brownian motions (SRBMs) constrained to orthants. The SRBMs in orthants arise as diffusion approximations for open queueing networks in heavy traffic; cf. Harrison and Williams (1987), Harrison and Nguyen (1993), and Williams (1996). Some important results on the existence, uniqueness, and positive recurrence of SRBMs can be found in the

survey paper by Williams (1995) (see our Section 2). In particular, Hobson and Rogers (1993) and Williams (1985) establish the sufficient and necessary condition for the positive recurrence of SRBMs in quadrants. A fluid-based sufficient condition for the positive recurrence of SRBMs in general dimensions is developed in Dupuis and Williams (1994), which is shown to be a necessary condition in three dimensions by Bramson et al. (2010). In the current work, we show that the fluid-based sufficient condition is also necessary for two dimensional SRBMs.

Since these papers, a vast literature on the topic has emerged, and of particular interest is the body of work concerned with the tail probabilities of stationary distributions of SRBMs (cf. Avram et al. 2001, Dupuis and Ramanan 2002, Dai and Miyazawa 2011). Until now, all results on tail probabilities have been for either two-dimensional systems or special cases of higher dimensional systems. For example, Liang and Hasenbein (2013) assume the property of rotational symmetry for three-dimensional SRBMs, and Dai et al. (2014) study how to ensure that the stationary distribution of an SRBM has a product form and how this can be used to identify the asymptotics of its tail probabilities. The work of El Kharroubi et al. (2012) analyzes the properties of the most likely trajectories that give rise to tail events in SRBMs. Given the challenge of developing analytical results for the tail probabilities of SRBMs, our work focuses on developing numerical algorithms to aid in the estimations of these probabilities.

Broadly speaking, there are two types of algorithms for estimating rare event probabilities: importance sampling and particle methods. The general idea of importance sampling is to simulate the system of interest under a new probability measure that makes the event of interest less rare and then adjust the estimator by the Radon–Nikodym derivative between the original measure and the sampling measure. In particle methods, one simulates multiple copies of the system of interest and uses some selection method to favor copies of the system in which the rare event of interest is more likely. In the current work, we focus on a particularly simple particle method called the “splitting algorithm.” In this method, one establishes landmarks in the state space, and each time a particle crosses a landmark, it is replaced with a random number of offspring. The creation of landmarks at which splitting events occur is, of course, the challenging aspect of creating efficient and stable splitting algorithms. Dean and Dupuis (2009) show how one can use a large deviation principle to ensure that the splitting algorithm is stable and efficiently estimates the rare event of interest. There have been several works (e.g., Dean and Dupuis 2009, 2011; Blanchet et al. 2011) that have looked at rare event simulations using particle methods, but to our knowledge, our work is the first to study rare event simulations for SRBMs using any method.

In this work, we develop splitting algorithms to estimate specific tail probabilities for SRBMs. We focus on the particular rare event that a positive recurrent SRBM exits the L^1 cube of radius n before returning to the cube of radius ϵ . The probability of such an event can be used to approximate the overflow probability that the total number of jobs in a queueing system reaches the level n during a busy period. We first show that, under a proper scaling and two regularity conditions, the probability of the rare event satisfies a large deviation principle. The two regularity conditions are the Lipschitz continuity of the Skorokhod problem (SP) associated with the SRBM (Assumption 1) and the positive recurrence of the SRBM. In particular, to ensure the positive recurrence property, we assume that the SRBMs satisfy the Dupuis–Williams stability condition (Assumption 2), which requires that the solution of a related SP is attracted to the origin.

The construction of a splitting algorithm is usually carried out by specifying a so-called *importance function* that specifies when splitting events occur. The innovative work of Dean and Dupuis (2009) shows that, if one uses an importance function that is a subsolution to the variational problem (VP) associated to the large deviation principle for the problem of interest, then it is possible to provide logarithmic bounds on the performance of the splitting algorithm. Dean and Dupuis (2009), in fact, show that, if one works with the maximal subsolution to the VP, then it is possible to guarantee subexponential growth in the computational complexity of the splitting algorithm as the event of interest becomes rarer, that is, an *asymptotically efficient* algorithm. Based on Dean and Dupuis (2009) our goal is to identify subsolutions to the VP associated with the rare event probability of interest. If possible, we would like to identify solutions to the VP because these are maximal subsolutions and thereby guarantee an asymptotically efficient algorithm. For positive recurrent, two-dimensional SRBMs, the VP has been exactly solved in Avram et al. (2001), and we are able to use this solution to build an importance function for two-dimensional SRBMs. It remains an open problem to solve the VP for SRBMs in three or higher dimensions (cf. El Kharroubi et al. 2012, Farlow 2013, Liang and Hasenbein 2013). The challenge in higher dimensional problems is that the optimal trajectory may contain an infinite number of piecewise linear components, which “spiral” around boundaries; see, for example, Dupuis and Ramanan (2002) and Avram et al. (2001). Because of the possibility of these paths being optimal, it is generally not possible to analytically solve the VP in dimensions higher than two.

For two-dimensional SRBMs, we show that the Dupuis–Williams stability condition is sufficient and necessary for the positive recurrence so that the exact solution to the VP from Avram et al. (2001) can be adapted directly without any additional conditions (Lemma 2). To study the VP in three and higher dimensions, we assume that

the reflection matrix R of the SRBM is a nonsingular $\mathcal{M}$ -matrix (Assumption 3). For the SRBM satisfying this assumption, it is shown that the Dupuis–Williams stability condition is also sufficient and necessary for its positive recurrence and is equivalent to the condition that $R^{-1}\theta < 0$, where θ is the drift of the SRBM (Lemma 3). Our novel approach for finding subsolutions to the VP in three and higher dimensions is to first establish in Lemma 4 that it suffices to find a function that satisfies the subsolution property for pairs of points constrained to a particular boundary face of $\mathbb{R}_+^d$; that is, it satisfies the subsolution property locally. We then use the results of El Kharroubi et al. (2012) to construct a local subsolution. The construction relies on both the $\mathcal{M}$ -matrix assumption and the equivalent stability condition $R^{-1}\theta < 0$ (see Lemma 6). We are then able to rescale and shift our local subsolution to ensure that it satisfies the necessary boundary conditions and to derive a subsolution to the VP (Theorem 2).

Given that it is not, in general, possible to exactly simulate an SRBM, we use the Euler method to simulate a discrete time approximation process. An important question about bias in our estimator then arises because we are, in fact, simulating a discretized approximation to the SRBM. We first establish conditions on the discretization step size to ensure the approximating process satisfies the same large deviation principle as the original SRBM (Proposition 4). With this information, we conclude that the logarithmic asymptotics of the variance of the estimators for the SRBM are the same as for estimating the rare event probability for the approximating process. A remaining issue that we need to address is to estimate the amount of bias present in our estimator resulting from the use of the discretized process. In Proposition 5, we establish bounds on the discretization bias of our estimator. In particular, we show that the growth of the relative bias term, as $n \rightarrow \infty$, is at most subexponential.

In summary, we show that our splitting algorithms can estimate the relevant rare events with asymptotic optimality for a discretization of two-dimensional SRBMs. For the three and higher dimensional case, although the proposed algorithm is not asymptotically optimal, it is stable and is exponentially superior to the standard Monte Carlo method. Specifically, the exponential decay rate of the variance of our estimator is strictly greater than that of the standard Monte Carlo estimator (see Section 5.4). The superior performance of our algorithms over standard Monte Carlo is verified numerically in Section 6. Using our results on the behavior of the discretized process, our splitting algorithms can be used for estimating the rare event probability for either SRBMs or the approximation processes with identical logarithmic asymptotics.

The remainder of this work is organized as follows. In Section 2, we define the SRBMs, state the basic assumptions, and define the rare event of interest. In Section 3, we show that the rare event probability satisfies a large deviation principle. The VPs for SRBMs are studied in Section 4, and a subsolution to the VP in general dimensions is constructed. In Section 5, we introduce the discretizations of SRBMs, apply the splitting method to the discretized process, and state our main results on the performance of the splitting method for estimating rare events. Section 6 presents numerical examples, Section 7 contains a summary of our results, and Section 8 provides the complete proofs of the major results. Finally, the appendix collects all other proofs and the analytical solution to the VP in two dimensions.

Throughout the manuscript, we use the following notation. The d -dimensional Euclidean space is denoted by $\mathbb{R}^d$, and its nonnegative orthant is denoted by $\mathbb{R}_+^d$, that is, $\mathbb{R}_+^d = \{x \in \mathbb{R}^d : x_i \geq 0, i = 1, \dots, d\}$. For a pair of d -dimensional vectors u and v , we denote their standard inner product by $\langle u, v \rangle = \sum_{i=1}^d u_i v_i$ and the standard Euclidean norm by $|u| = \sqrt{\langle u, u \rangle}$. We write $u \leq v$ if $u_i \leq v_i$ for $i = 1, \dots, d$. For a positive definite $d \times d$ matrix A and d -dimensional vectors u and v , define the associated inner product by $\langle u, v \rangle_A = \langle u, Av \rangle$ and the associated norm $|u|_A = \sqrt{\langle u, u \rangle_A}$. The trace of a square matrix A is denoted by $\text{tr}(A)$, and the transpose of a matrix B is denoted by B^T . For a Polish space S , denote by $\mathcal{C}([0, \infty); S)$ the space of continuous functions from $[0, \infty)$ to S , $\mathcal{AC}([0, \infty); S)$ the space of absolutely continuous functions from $[0, \infty)$ to S , and $\mathcal{C}^2([0, \infty); S)$ the space of twice differentiable functions from $[0, \infty)$ to S . For $f \in \mathcal{C}^2([0, \infty); S)$, let $\dot{f}$ denote its gradient vector and $\ddot{f}$ its Hessian matrix. For a Borel set $B \subset \mathbb{R}^d$, and a constant $c \in \mathbb{R}$, $cB = \{cz \in \mathbb{R}^d : z \in B\}$. For a set K of a finite number of elements, denote by $\text{card}(K)$ its cardinality. Throughout the paper, we use the following notation for the asymptotic behavior of positive functions:

$$\begin{aligned} f(t) &\sim g(t) && \text{if } f(t)/g(t) \rightarrow 1 \text{ as } t \rightarrow \infty, \\ f(t) &= o(g(t)) && \text{if } f(t)/g(t) \rightarrow 0 \text{ as } t \rightarrow \infty, \\ f(t) &= O(g(t)) && \text{if } f(t) \leq Cg(t) \text{ for all } t, \\ f(t) &= \Theta(g(t)) && \text{if } cg(t) \leq |f(t)| \leq Cg(t) \text{ for all } t, \end{aligned}$$

where C and c are positive constants.

2. Problem Formulation

The SRBMs in the nonnegative orthant were first introduced in Harrison and Reiman (1981). Roughly speaking, such a diffusion process behaves like a Brownian motion in the interior of the orthant. Upon reaching the boundary, it is reflected back into the interior in some specified direction so as to constrain the process in the orthant. The rigorous definition of SRBMs is stated as follows.

Definition 1 (SRBM). An SRBM associated with data (θ, Σ, R) in $\mathbb{R}_+^d$ is a continuous $\{\mathcal{F}_t\}$ -adapted d -dimensional process Z defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ such that

$$Z(t) = z + X(t) + \theta t + RY(t) \geq 0, \text{ for } t \in \mathbb{R}_+^d, \quad (2.1)$$

where $z \in \mathbb{R}_+^d$, X is a d -dimensional driftless Brownian motion with covariance matrix Σ and $X(0) = 0$, and Y is a d -dimensional stochastic process such that, for $i = 1, \dots, d$,

- i. $Y_i(0) = 0$;
- ii. Y_i is continuous and nondecreasing;
- iii. Y_i can increase only when Z_i is on the set $F_i = \{x \in \mathbb{R}_+^d : x_i = 0\}$, that is, $\int_0^\infty Z_i(t) dY_i(t) = 0$.

The set F_i is called the i th face of the orthant $\mathbb{R}_+^d$. The parameters θ, Σ , and R are known as the drift, covariance matrix, and reflection matrix of the SRBM. We assume that Σ is invertible.

The SRBM Equation (2.1) admits a weak solution if there exist a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ and the process (X, Y, Z) that satisfies (2.1) (cf. Taylor and Williams 1993). It is shown in Taylor and Williams (1993) that there exists a (weak) SRBM Z associated with (θ, Σ, R) if and only if R is completely $\mathcal{S}$, and in this case, Z is unique in law and is a Feller continuous strong Markov process. A matrix is said to be completely $\mathcal{S}$ if, for its every $k \times k$ principle submatrix G , there exists a k -dimensional vector $v_G \geq 0$ such that $Gv_G > 0$.

An important tool in the study of SRBMs is the SP. Let $\mathcal{C}_0([0, \infty); \mathbb{R}^d) = \{f \in \mathcal{C}([0, \infty); \mathbb{R}^d) : f(0) \in \mathbb{R}_+^d\}$.

Definition 2 (SP). Let $\psi \in \mathcal{C}_0([0, \infty); \mathbb{R}^d)$. A pair $(\phi, \eta) \in \mathcal{C}([0, \infty); \mathbb{R}_+^d) \times \mathcal{C}([0, \infty); \mathbb{R}_+^d)$ is called a solution of the SP for ψ associated with R if the following hold:

- i. For all $t \geq 0$, $\phi(t) = \psi(t) + R\eta(t) \geq 0$.
- ii. The function η satisfies the following: (a) $\eta_i(0) = 0$, (b) $\eta_i(\cdot)$ is nondecreasing, and (c) $\eta_i(\cdot)$ increases only when $\phi_i(\cdot) = 0$, that is, $\int_0^\infty \phi_i(t) d\eta_i(t) = 0$.

Theorem 2.2 in Williams (1995) guarantees that there is a solution to the SP for ψ if and only if R is completely $\mathcal{S}$. Define $\Gamma : \mathcal{C}_0([0, \infty); \mathbb{R}^d) \rightarrow \mathcal{C}([0, \infty); \mathbb{R}_+^d); \psi \mapsto \phi$.

Definition 3 (Skorokhod map (SM) and R -regulation). If $\Gamma(\psi)$ is unique for each $\psi \in \mathcal{C}_0([0, \infty); \mathbb{R}^d)$, then Γ is called the SM associated with the matrix R , and we call $\Gamma(\psi)$ the R -regulation for ψ .

Throughout this work we make the following assumption to ensure that our SRBM satisfies a sample path large deviation principle.

Assumption 1. The SM Γ is well defined on $\mathcal{C}_0([0, \infty); \mathbb{R}^d)$, and is Lipschitz continuous in the following sense: there exist $\kappa > 0$ so that, for $\psi_1, \psi_2 \in \mathcal{C}_0([0, \infty); \mathbb{R}^d)$ and $T > 0$,

$$\sup_{t \in [0, T]} |\Gamma(\psi_1)(t) - \Gamma(\psi_2)(t)| \leq \kappa \sup_{t \in [0, T]} |\psi_1(t) - \psi_2(t)|.$$

Note that Assumption 1 implies that R is completely $\mathcal{S}$. Some sufficient conditions for the Lipschitz continuity of Γ can be found in Harrison and Reiman (1981), Dupuis and Ishii (1991), and Dupuis and Ramanan (1999). In particular, if R is a generalized $\mathcal{M}$ -matrix, then Γ is Lipschitz continuous. Here R is a generalized $\mathcal{M}$ -matrix if (i) the diagonal entries of R are positive, and (ii) writing $R = (\mathbb{I}_d - Q)\text{diag}(R)$, where $\text{diag}(R)$ is a diagonal matrix with the i th diagonal entry equal to R_{ii} , $|Q|$ (obtained by replacing all the entries of Q by their absolute values) has special radius less than one. This sufficient condition is referred to as the *generalized Harrison–Reiman condition* as in Dupuis and Ishii (1991) and other work, and the original Harrison and Reiman condition established in Harrison and Reiman (1981) requires that Q has nonnegative entries. The definition of $\mathcal{M}$ -matrices is given in Section 4 preceding Assumption 3 and is, in fact, equivalent to the original Harrison–Reiman condition (see Lemma 3 and its proof).

We require that the SRBM Z is positive recurrent so that the expected time that Z reaches any closed set with positive Lebesgue measure is finite.

Definition 4 (Positive Recurrence). The SRBM Z is positive recurrent if, for every closed set F in $\mathbb{R}_+^d$ with positive Lebesgue measure, $\mathbb{E}[\tau_F] < \infty$, where $\tau_F = \inf\{t \geq 0 : Z(t) \in F\}$.

Based on theorem 2.6 of Dupuis and Williams (1994), we make the following assumption that provides a sufficient condition for the positive recurrence of SRBMs. This assumption is referred to as the Dupuis–Williams stability condition and is shown to be sufficient and necessary for the positive recurrence of the SRBMs considered for the VP in Section 4 (see Lemmas 2 and 3).

Assumption 2 (Dupuis–Williams). Let $\psi \in \mathcal{C}_0([0, \infty); \mathbb{R}^d)$ be of the form $\psi(t) = z + \theta t$, $t \geq 0$. Then, $\lim_{t \rightarrow \infty} \Gamma(\psi)(t) = 0$.

For a positive recurrent SRBM Z , we introduce a large deviation scaling with the scaling parameter $n \in \mathbb{N}$ as follows: for $n \in \mathbb{N}$, define

$$Z_n(t) = \frac{Z(nt)}{n}, \quad t \geq 0.$$

The process Z_n starts in a point $z_n \notin A_n \cup B$. We assume $z_n \rightarrow 0$ in our main results. This says the initial value z of the unscaled SRBM Z is $o(n)$ under the large deviation setting. Let $\epsilon \in (0, 1)$. Consider two sets $A_n = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \epsilon/n\}$ and $B = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq 1\}$. We are interested in the probability that the process Z_n first enters B before reaching A_n with a starting point $z_n \notin A_n \cup B$. Letting $\tau_n = \inf\{t \geq 0 : Z_n(t) \in A_n \cup B\}$, the probability of interest can be defined as

$$p_n(z_n) = \mathbb{P}(Z_n(\tau_n) \in B | Z_n(0) = z_n). \quad (2.2)$$

The SRBMs in nonnegative orthants arise as diffusion approximations for open queueing networks in heavy traffic. The probability considered here corresponds to the overflow probability that the total number of customers in the queueing system reaches level n within a busy period.

The goal of this work is to build algorithms that can stably estimate the probability $p_n(z_n)$ and provide an improvement over the standard Monte Carlo. Given the positive recurrence assumption on the SRBM Z , it is intuitively obvious that p_n goes to zero as $n \rightarrow \infty$. Furthermore, the exponential decay rate of the probability is clearly related to the large deviation properties of the sequence $\{Z_n\}_{n \geq 1}$. One minor point that we have to deal with is that the sets $\{A_n\}_{n \geq 1}$ are dependent on the scaling factor n . Ideally, we would like to replace the sequence of sets $\{A_n\}_{n \geq 1}$ with the set $\{0\}$; however, for positive recurrent SRBMs, it is possible that the expected time to hit point 0 is infinite. In the next section, we develop the necessary large deviation theory to characterize (in terms of a variational problem) the exponential decay rate of $p_n(z_n)$.

3. Large Deviation Principle

For $T \geq 0$, using Schilder's theorem and the contraction principle, $\{Z_n(t); t \in [0, T]\}$ with initial value $Z_n(0) = z$ satisfies the large deviation principle (LDP) in $\mathcal{C}([0, T]; \mathbb{R}_+^d)$ with the good rate function

$$I_{T,z}(\phi) = \inf_{\psi \in \mathcal{AC}_z([0, T]; \mathbb{R}_+^d), \phi = \Gamma_T(\psi)} \frac{1}{2} \int_0^T |\dot{\psi}(t) - \theta|_{\Sigma^{-1}}^2 dt, \quad (3.1)$$

where $\Gamma_T(\psi)$ is the R -regulation for $\{\psi(t); t \in [0, T]\}$, and

$$\mathcal{AC}_z([0, T]; \mathbb{R}_+^d) = \{\psi \in \mathcal{AC}([0, T]; \mathbb{R}_+^d) : \psi(0) = z\}.$$

In particular, we have the following result for any $T > 0$. Its proof is a simple application of the contraction principle and is, thus, omitted.

Lemma 1. Assume that $z_n \rightarrow z \in \mathbb{R}_+^d$ and let $I_{T,z}$ be as in (3.1). Then, for any closed $F \subset \mathcal{C}([0, T]; \mathbb{R}_+^d)$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(Z_n \in F) \leq - \inf_{\phi \in F} I_{T,z}(\phi);$$

for any open $G \subset \mathcal{C}([0, T]; \mathbb{R}_+^d)$,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(Z_n \in G) \geq - \inf_{\phi \in G} I_{T,z}(\phi).$$

Note that the probability $p_n(z_n)$ is time independent, and we need to extend the previous result to handle infinite time horizon events. A standard technique is to use the stability properties of the stochastic process to establish an upper bound on the size of the stopping time τ_n . Recall that $\tau_n = \inf\{t \geq 0 : Z_n(t) \in A_n \cup B\}$, where $A_n = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \epsilon/n\}$ for $\epsilon \in (0, 1)$, and $B = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq 1\}$. The result is present in the following proposition.

Proposition 1. Assume that $z_n \rightarrow z \in \mathbb{R}_+^d$. Under Assumptions 1 and 2, there exists $c \in (0, \infty)$ such that

$$\limsup_{n \rightarrow \infty} \sup_{z \in \mathbb{R}_+^d} \frac{1}{n} \log \mathbb{E}_{z_n} (e^{cn\tau_n}) < \infty. \quad (3.2)$$

Here, $\mathbb{E}_{z_n}$ denotes the expectation conditioning on $Z_n(0) = z_n$. If z_n further satisfies that $z_n = O(1/n)$, then for some $c_1, c_2 \in (0, \infty)$,

$$\mathbb{P}_{z_n}(n\tau_n > t) \leq c_1 e^{-c_2 t}.$$

Outline of Proof. The complete proof of Proposition 1 is provided in Section 8.1. Define $\tau_\epsilon = \inf \{t \geq 0 : \sum_{i=1}^d Z_i(t) \leq \epsilon\} = n \inf \{t \geq 0 : Z_n(t) \in A_n\}$, where Z is the unscaled SRBM, and then, $n\tau_n \leq \tau_\epsilon$. We show (3.2) with $n\tau_n$ replaced by τ_ϵ and $z \in \mathbb{R}_+^d$ replaced by $z \in \mathbb{R}_+^d \setminus B$. We apply the Lyapunov function $L : \mathbb{R}_+^K \rightarrow \mathbb{R}_+$, which is introduced in Dupuis and Williams (1994), to the SRBM Z , and consider the Ito formula of $L(Z(t))$. Using the nice properties of L (see Theorem 3), the probability $\mathbb{P}_{z_n}(\tau_n > t)$ can be bounded above by a probability related to the martingale $\int_0^t \langle \dot{L}(Z(u)), dX(u) \rangle$. The latter can be estimated to have an upper bound that decays exponentially in t . The result then follows. $\square$

We now consider an infinite horizon variational problem that minimizes over all paths that travel from z to a given point x . In particular, define the quasipotential

$$I_z(x) = \inf_{T \geq 0} \inf_{\psi \in \mathcal{AC}_z([0, \infty); \mathbb{R}^d), \Gamma(\psi)(T) = x} \frac{1}{2} \int_0^T |\dot{\psi}(t) - \theta|_{\Sigma^{-1}}^2 dt. \quad (3.3)$$

With Proposition 1, the finite horizon LDP can be extended to infinite time horizon events and used to study $p_n(z_n)$. The following theorem establishes the LDP for $p_n(z_n)$ when $z_n \rightarrow 0$. We let $I(x) = I_0(x)$ for $x \in \mathbb{R}_+^d$.

Theorem 1. Assume $z_n \in \mathbb{R}_+^d \setminus (A_n \cup B)$ for each n , and $z_n \rightarrow 0$. Under Assumptions 1 and 2,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) = - \inf_{x \in B} I(x).$$

Outline of Proof. To prove Theorem 1, we consider the following two inequalities separately.

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \leq - \inf_{x \in B} I(x), \quad (3.4)$$

and

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \geq - \inf_{x \in B} I(x). \quad (3.5)$$

The upper bound Inequality (3.4) follows from Proposition 1 and a standard argument similar to lemma 5.7.21 in Dembo and Zeitouni (2010). To show the lower bound Inequality (3.5), it suffices to establish that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \geq - \frac{1}{2} \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds$$

for any $\psi \in \mathcal{AC}([0, \infty), \mathbb{R}^d)$ such that $\psi(0) = 0$ and $\tau_\psi < \infty$. We first shift ψ by a constant and define $\varphi^x = \psi + x$, where $x \in B_\kappa = \{x \in \mathbb{R}_+^d : \sum_{i=1}^d x_i = \kappa\}$ with $\kappa \in (0, 1)$. For each ψ , we construct an open set consisting of functions $\eta \in \mathcal{C}([0, \tau_\psi]; \mathbb{R}_+^d)$ that are close to $\Gamma(\varphi^x)$. Next, using the strong Markov property, we decompose $p_n(z_n)$ as follows.

$$p_n(z_n) \geq \mathbb{P}_{z_n}(Z_n \text{ hits } B_\kappa \text{ before hitting } A_n) \mathbb{P}(Z_n \text{ hits } B \text{ before hitting } A_n | Z_n(0) \in B_\kappa).$$

The first probability in the right-hand side can be estimated using the minimality property of the one-dimensional Skorokhod map, and the second probability can be analyzed using the open set defined for ψ . The complete proof can be found in Section 8.2. $\square$

Remark 1. Because of the Dupuis–Williams assumption, we expect that, when $z_n \rightarrow z \neq 0$ and $z \notin B$, a result similar to Theorem 1 still holds for $p_n(z_n)$. However, because the limit point $z \neq 0$, the variational problem governing

the logarithmic asymptotics of $p_n(z_n)$ are more complicated. In particular, the variational problem has to exclude sample paths that return to the radius ϵ ball around the origin. In order to streamline our arguments and the later construction of our simulation algorithm, we focus in the remainder of the manuscript on the setting in which $z_n \rightarrow 0$.

4. Variational Problems

This section focuses on the study of the VP defined in (3.3) when $z = 0$. A triplet (ψ, η, ϕ) is called a solution to the VP (3.3) if (ϕ, η) is a solution of the SP for ψ , $\phi(T) = z$ for some $T \geq 0$, and

$$\frac{1}{2} \int_0^T |\dot{\psi}(t) - \theta|_{\Sigma^{-1}}^2 dt = I(z).$$

The solution (ψ, η, ϕ) is also referred to as an *optimal triple*, and ψ and ϕ are referred to as *optimal path* and *optimal reflected path*, respectively.

4.1. The VP for Two-Dimensional SRBMs

It is known from Avram et al. (2001) that the VP in $\mathbb{R}_+^2$ admits an explicit analytical solution (we summarize their solution construction in Section A.2). Our simulation algorithms developed in Section 5 are based on this solution in the case of two dimensions (see (5.6)). In Avram et al. (2001), the VP is studied under the stability condition that

$$\theta_1 + r_2 \theta_2^- < 0, \quad \theta_2 + r_1 \theta_1^- < 0, \quad (4.1)$$

where r_1 and r_2 are the off-diagonal entries of R , and R is assumed to be of the following form:

$$R = \begin{bmatrix} 1 & r_2 \\ r_1 & 1 \end{bmatrix}. \quad (4.2)$$

It is well known that Condition (4.1) is sufficient and necessary for the positive recurrence of two-dimensional SRBMs (see section 4.1 in Avram et al. 2001). Also, from Harrison and Hasenbein (2009), Condition (4.1) is equivalent to the following two conditions

$$R \text{ is a } \mathcal{P} \text{ matrix, that is } R \text{ has a positive determinant,} \quad (4.3)$$

$$R^{-1} \theta < 0, \text{ which says } \theta_1 - r_2 \theta_2 < 0, \theta_2 - r_1 \theta_1 < 0. \quad (4.4)$$

In the next lemma, we show that, in two dimensions, Condition (4.1) is equivalent to the Dupuis–Williams stability condition (Assumption 2). Therefore, we can use the results in Avram et al. (2001) under Assumption 2 and don't need any further assumptions.

Lemma 2. *For two-dimensional SRBMs, the Dupuis–Williams stability condition (Assumption 2) is equivalent to the stability condition (4.1). Consequently, the Dupuis–Williams stability condition (Assumption 2) is both sufficient and necessary for the positive recurrence of the two-dimensional SRBMs.*

Proof. It suffices to show that Condition (4.1) implies the Dupuis–Williams stability condition (Assumption 2). The following proof is similar to that of lemma 14 in Bramson et al. (2010). Noting that R^{-1} is also a $\mathcal{P}$ matrix, there exists $(d_1, d_2) > 0$ such that $(c_1, c_2) = (d_1, d_2)R^{-1} > 0$. We observe from (4.4) that

$$c_1 \theta_1 + c_2 \theta_2 = (d_1, d_2)R^{-1} \theta < 0.$$

Now, consider $\psi(t) = z + \theta t$ and its R -regulation $\phi(t)$. From the oscillation property (see Williams 2017), both ϕ and η are Lipschitz continuous. Define $g(t) = c_1 \phi_1(t) + c_2 \phi_2(t)$. To show that $\phi(t) \rightarrow 0$ as $t \rightarrow \infty$, it suffices to show that $\dot{\phi}(t) \leq c$ for some $c < 0$ almost everywhere t . When $\phi_1(t) > 0$ and $\phi_2(t) > 0$, $\dot{\phi}_1(t) = \theta_1$ and $\dot{\phi}_2(t) = \theta_2$. Thus, $\dot{g}(t) = c_1 \theta_1 + c_2 \theta_2 < 0$. Suppose $\phi_1(t) = 0$ and $\phi_2(t) > 0$ for $t \in [t_1, t_2]$. For $t \in (t_1, t_2)$, we have $\dot{\phi}_1(t) = 0$ and $\dot{\eta}_2(t) = 0$, which yields that $\dot{\eta}_1(t) = -\theta_1$, and from (4.4), $\dot{\phi}_2(t) = \theta_2 - r_1 \theta_1 < 0$, and further $\dot{g}(t) = c_2(\theta_2 - r_1 \theta_1) < 0$. Similarly, if $\phi_1(t) > 0$ and $\phi_2(t) = 0$ for $t \in [t_1, t_2]$, then for $t \in (t_1, t_2)$, $\dot{g}(t) = c_1(\theta_1 - r_2 \theta_2) < 0$. $\square$

For the Lipschitz continuity of Skorokhod problems associated with R defined in (4.2), the generalized Harrison–Reiman condition requires that $|r_1 r_2| < 1$, which certainly implies Condition (4.3). Consequently, a sufficient condition for both the Lipschitz continuity (Assumption 1) of Skorokhod problems in two dimensions and the positive recurrence (Assumption 2) of two-dimensional SRBMs is given:

$$|r_1 r_2| < 1, \text{ and } R^{-1} \theta < 0. \quad (4.5)$$

4.2. The VP for Three and Higher Dimensional SRBMs

For the VP in $\mathbb{R}_+^3$, a special case of a rotationally symmetric VP is studied in Liang and Hasenbein (2013), and general paths properties are studied in El Kharroubi et al. (2012). In Farlow (2013), the value of the VP in $\mathbb{R}_+^d$ is characterized as a solution to a partial differential equation. However, the complete analysis of the VP in three or higher dimensional space is still open. In the following, we apply the path properties from El Kharroubi et al. (2012) to construct a subsolution for the VP in $\mathbb{R}_+^d$ when $d \geq 3$.

We further assume that the reflection matrix R is an $\mathcal{M}$ -matrix. A $d \times d$ matrix is called a $\mathcal{Z}$ -matrix if its off-diagonal entries are nonpositive, and a $\mathcal{Z}$ -matrix is a nonsingular $\mathcal{M}$ -matrix if it can be expressed as $\mu \mathbb{I}_d - A$, where $\mathbb{I}_d$ is a $d \times d$ identity matrix, A is a $d \times d$ matrix with nonnegative entries, and μ is greater than the spectral radius of A . Formally, we make the following assumption on R .

Assumption 3 ($\mathcal{M}$ -matrix). *The reflection matrix R is a nonsingular $\mathcal{M}$ -matrix.*

SRBMs with reflection matrix R , that is, an $\mathcal{M}$ -matrix, arise as diffusion approximations for single class open queueing networks in heavy traffic (see Reiman 1984). This $\mathcal{M}$ -matrix assumption is, in fact, equivalent to the original Harrison–Reiman condition (see the proof of Lemma 3 for more details), and thus, under Assumption 3, the Skorokhod problem is Lipschitz continuous. We further note that, when R is a nonsingular $\mathcal{M}$ -matrix, the SRBM Z is positive recurrent if and only if $R^{-1}\theta < 0$ (cf. Harrison and Williams 1987). The following lemma shows that, under Assumption 3, the Dupuis–Williams stability condition (Assumption 2) is equivalent to the condition that $R^{-1}\theta < 0$, and also summarizes the Lipschitz continuity of the SP under Assumption 3.

Lemma 3. *For SRBMs satisfying Assumption 3, the Dupuis–Williams stability condition (Assumption 2) is equivalent to the condition $R^{-1}\theta < 0$. Consequently, under Assumption 3, the Dupuis–Williams stability condition (Assumption 2) is sufficient and necessary for the positive recurrence of the SRBMs. Furthermore, under Assumption 3, the Skorokhod problem is Lipschitz continuous.*

Proof. It suffices to show that, if $R^{-1}\theta < 0$, then for $\psi \in C_0([0, \infty); \mathbb{R}^d)$ of the form $\psi(t) = z + \theta t$, $t \geq 0$, the R -regulation ϕ satisfies that $\lim_{t \rightarrow \infty} \phi(t) = 0$. We construct an SP with normal reflection that can dominate the original SP. The construction idea goes back to Harrison and Williams (1987). Let $v > 0$ in $\mathbb{R}^d$ satisfy $R^{-1}v \leq -R^{-1}\theta$. Define

$$\tilde{\phi}(t) = z - vt + \tilde{\eta}(t),$$

where $(\tilde{\phi}, \tilde{\eta})$ is the unique solution to the SP for $\tilde{\psi}(t) \equiv z - vt$, $t \geq 0$, associated with the identity reflection matrix. We next observe that

$$\tilde{\phi}(t) = z + \theta t + R\eta^*(t),$$

where $\eta^*(t) = R^{-1}\tilde{\eta}(t) + (-R^{-1}\theta - R^{-1}v)t$ is a nondecreasing function. Now, noting that $\phi(t) = z + \theta t + R\eta(t)$ and using the minimality property of the Skorokhod problem associated with R (see Reiman 1984), it follows that, for $t \geq 0$,

$$\eta^*(t) \geq \eta(t),$$

which yields that

$$R^{-1}\phi(t) \leq R^{-1}\tilde{\phi}(t), \quad t \geq 0.$$

Because R is an $\mathcal{M}$ -matrix, R^{-1} has positive diagonal entries and nonnegative off-diagonal entries. So, to show $\phi(t) \rightarrow 0$ as $t \rightarrow \infty$, it suffices to show that $\tilde{\phi}(t) \rightarrow 0$ as $t \rightarrow \infty$. Considering each component of $\tilde{\phi}$, we have for $i = 1, \dots, d$, $(\tilde{\phi}_i, \tilde{\eta}_i)$ is the solution to the one-dimensional SP, and

$$\tilde{\phi}_i(t) = z_i - v_i t + \sup_{s \in [0, t]} (-z_i + v_i s)^+ = 0, \quad t \geq z_i / v_i.$$

At last, we show that, for a nonsingular matrix R , if R is an $\mathcal{M}$ -matrix, then it is a generalized $\mathcal{M}$ -matrix, and if it is a generalized $\mathcal{M}$ -matrix with a nonnegative Q , then it is an $\mathcal{M}$ -matrix. Assume that R is a nonsingular $\mathcal{M}$ -matrix. Then $R \text{diag}(R)^{-1}$ is also an $\mathcal{M}$ -matrix. Thus, for some constant $\tilde{\mu} > 0$ and $d \times d$ matrix $\tilde{B} \geq 0$ whose spectral radius is less than $\tilde{\mu}$, we have

$$R \text{diag}(R)^{-1} = \tilde{\mu} \mathbb{I}_d - \tilde{B} = \mathbb{I}_d - (\tilde{B} - (\tilde{\mu} - 1) \mathbb{I}_d). \quad (4.6)$$

We note that $\tilde{R}$ has diagonal entries equal to one, which yields that the diagonal entries of $\tilde{B}$ equal to $\tilde{\mu} - 1$. Because $\tilde{B} \geq 0$, we have $\tilde{\mu} - 1 \geq 0$. Furthermore, the eigenvalues of $\tilde{B} - (\tilde{\mu} - 1) \mathbb{I}_d$ are equal to the eigenvalues of $\tilde{B}$

minus $(\tilde{\mu} - 1)$. Now, from the Perron–Frobenius theorem, both $\tilde{B}$ and $\tilde{B} - (\tilde{\mu} - 1)\mathbb{I}_d$ have positive eigenvalues that are greater than the moduli of all other eigenvalues; that is, the spectral radii are given by these positive eigenvalues. It follows now that the spectral radius of $\tilde{B} - (\tilde{\mu} - 1)\mathbb{I}_d$ is less than one, and from (4.6), R is a generalized $\mathcal{M}$ -matrix. Now suppose R is a generalized $\mathcal{M}$ -matrix with nonnegative Q of spectral radius less than one, that is, $R = (\mathbb{I}_d - Q)\text{diag}(R)$. Clearly, $\mathbb{I}_d - Q$ is an $\mathcal{M}$ -matrix, and so is R . This completes the proof. $\square$

We now study the VP in d dimensions, where $d \geq 3$. First, set $L = \{1, 2, \dots, d\}$. For $J, K \subset L$, define R_{JK} to be a $\text{card}(J) \times \text{card}(K)$ matrix by deleting rows and columns with indices in $L \setminus J$ and $L \setminus K$, respectively. Let R_j denote the j th column of R . Finally, define the face associated with $K \subset L$ to be $F_K = \{x \in \mathbb{R}_+^d : x_i = 0, i \in K, x_j > 0, j \in L \setminus K\}$.

For any $w, v \in \mathbb{R}_+^d, w \neq v$, define the globally optimal cost from w to v as follows:

$$\mathcal{I}(w, v) = \inf_{\psi \in \mathcal{AC}_w([0, \infty); \mathbb{R}^K)} \inf_{\phi = \Gamma(\psi), \tau_\phi(v) < \infty} \frac{1}{2} \int_0^{\tau_\phi(v)} |\dot{\psi}(t) - \theta|_{\Sigma^{-1}}^2 dt, \quad (4.7)$$

where

$$\tau_\phi(v) = \inf \{t \geq 0 : \phi(t) = v\}.$$

A path ψ^* that leads to the globally optimal cost is called a globally optimal path from w to v . We next define the locally optimal cost between points w and v , where we constrain the path to lie on a face F_K . In particular, we consider the local cost between points satisfying the following condition with respect to $K \subset L$. For $w, v \in \mathbb{R}_+^d$, define $w \vee v = (w_1 \vee v_1, w_2 \vee v_2, \dots, w_d \vee v_d)'$.

Condition 1. $w \neq v, w \vee v \in F_K$.

For the pair v and w satisfying Condition 1 with respect to $K \subset L$, we define the locally optimal cost from w to v with path constrained in F_K as follows:

$$\mathcal{I}_K^*(w, v) = \inf_{\psi \in \mathcal{AC}_w([0, \infty); \mathbb{R}^K)} \inf_{\phi = \Gamma(\psi), \phi(t) \in F_K, \forall t \in (0, \tau_\phi(v)), \tau_\phi(v) < \infty} \frac{1}{2} \int_0^{\tau_\phi(v)} |\dot{\psi}(t) - \theta|_{\Sigma^{-1}}^2 dt. \quad (4.8)$$

We now introduce the definition of a subsolution to the VP (3.3).

Definition 5. A function $\tilde{T} : \mathbb{R}_+^d \rightarrow \mathbb{R}$ is called a subsolution to the VP (3.3) if

- i. $\tilde{T}$ is continuous;
- ii. $\tilde{T}(x) \leq 0$ for all $x \in B$;
- iii. $\tilde{T}(w) - \tilde{T}(v) \leq \mathcal{I}(w, v)$ for all $w, v \in \mathbb{R}_+^d \setminus (\{0\} \cup B)$.

One method of identifying the subsolution $\tilde{T}$ to the VP is to identify viscosity subsolutions to a related partial differential equation (PDE). This topic is discussed in section 5 of Dean and Dupuis (2009); also see Budhiraja and Dupuis (2019) for many examples of the use of viscosity (and smooth) subsolutions in rare event analysis problems. This connection to PDEs is standard for VPs and follows by using the dynamic programming principle. However, because of the presence of the reflection map, this process does not work well in the current problem. In particular, as noted, for example, in Dupuis and Ramanan (2002), because of the reflection map, the independent variable in $V(x)$ is the terminal position of the trajectory. As a result, it is nontrivial to identify the correct structure of the related PDE along the boundaries of the orthant. In Farlow and Day (2015), they identify necessary boundary conditions, but these conditions are rather difficult to work with along lower dimensional boundaries. Given the difficulty of the PDE in higher dimensions, we work directly with the VP.

The following lemma is a key step in our construction of a subsolution.

Lemma 4. Let $\tilde{T}$ be a function that satisfies conditions (i) and (ii) in Definition 5. If, in addition, for all $K \subseteq L$, $\tilde{T}(w) - \tilde{T}(v) \leq \mathcal{I}_K^*(w, v)$ for all w, v satisfying Condition 1 with respect to K , then it is a subsolution to the VP (3.3).

Proof. For any $w, v \in \mathbb{R}_+^d \setminus (\{0\} \cup B)$, suppose we have an optimal triple (ψ, η, ϕ) with $\phi(0) = w, \phi(T) = v$. Recall that ϕ is continuous and of bounded variation (because ψ is absolutely continuous, η is continuous and monotone). Denote by $\mathcal{P}(I)$ the power set of L . When ϕ traverses a face F_K from time t_1 to t_2 for some $K \in \mathcal{P}(I)$, the segment of (ψ, η, ϕ) must be locally optimal. Otherwise, we can find a path that has lower cost. For each $K \in \mathcal{P}(I)$, we define the set

$$G_K = \{t \in [0, T] : \phi_i(t) = 0, \phi_j(t) > 0, \forall i \in K, \forall j \in L \setminus K\}^o,$$

where for a set E , E^o denotes its interior. We notice that, for each K , G_K is an open set and can be represented as the disjoint union of at most countably many open intervals. Denote the collection of these open intervals by $\mathcal{O}_K$.

Notice that the optimal path traverses the face F_K from time t_1 to t_2 if $(t_1, t_2) \in \mathbb{O}_K$. Observe that $\bigcup_{K \in \mathcal{P}(I)} \mathbb{O}_K$ contains at most countably many disjoint open intervals, and

$$\begin{aligned} \mathcal{I}(w, v) &\geq \sum_{K \in \mathcal{P}(I)} \sum_{O \in \mathbb{O}_K} \mathcal{I}_K^*(\phi(\inf(O)), \phi(\sup(O))) \\ &\geq \sum_{K \in \mathcal{P}(I)} \sum_{O \in \mathbb{O}_K} \bar{T}(\phi(\inf(O))) - \bar{T}(\phi(\sup(O))) \\ &= \bar{T}(w) - \bar{T}(v), \end{aligned}$$

Because $\bar{T}$ is continuous, the last equality can be justified by the facts that (1) there are at most countably many singleton points in the set $[0, T] \setminus \{\bigcup_{K \in \mathcal{P}(I)} G_K\}$ (otherwise, it violates the fact that ϕ is of bounded variation), and (2) ϕ is continuous. $\square$

A local VP similar to the VP (4.8) has been investigated in El Kharroubi et al. (2012). Specifically, two types of locally optimal paths were considered: interior escape paths and single segment boundary escape paths. An interior escape path is the one for which no reflection is applied. Let K and K' be two subsets of L such that $K \subset K'$ and $0 < \text{card}(K) < \text{card}(K') \leq d$. For $w \in F_{K'}$, $v \in F_K$, an optimal single segment boundary escape path is a path constrained in F_K that has the cost equal to $\mathcal{I}_K^*(w, v)$. The result in El Kharroubi et al. (2012) can be applied to the VP (4.8) directly, and their proofs are still valid for our setting.

In order to state the results in El Kharroubi et al. (2012) on $\mathcal{I}_K^*(\cdot, \cdot)$, we need some further notation. Given $K \subseteq L$ and two points $v, w \in \mathbb{R}_+^d$ satisfying Condition 1 with respect to K , for each $J \subseteq K$, define

$$\begin{aligned} B^J &= (R_{LJ}^T \Sigma^{-1} R_{LJ})^{-1} R_{LJ}^T \Sigma^{-1}, \text{ for } J \neq \emptyset, A^J \\ &= \begin{cases} \mathbb{I}_d - R_{LJ} B^J, & \text{for } J \neq \emptyset, \\ \mathbb{I}_d, & \text{for } J = \emptyset, \end{cases} \\ \alpha^J(w, v) &= \frac{|A^J \theta|_{\Sigma^{-1}}}{|A^J(v-w)|_{\Sigma^{-1}}}, \\ \lambda^J(w, v) &= \frac{|A^J \theta|_{\Sigma^{-1}}}{|A^J(v-w)|_{\Sigma^{-1}}} B^J(v-w) - B^J \theta, \text{ for } J \neq \emptyset. \end{aligned}$$

The following Proposition 2 summarizes some major results of El Kharroubi et al. (2012), including theorems 3 and 4, lemma 5, and propositions 1 and 2. Despite a small difference in problem settings, their proofs are still valid for our setting (see appendix for a complete proof).

Proposition 2. *Given $K \subseteq L$, and two points $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1 with respect to K , there exists a unique $J_* \subseteq K$ such that $\langle A^{J_*} R_j, \alpha^{J_*}(w, v)(v-w) - \theta \rangle_{\Sigma^{-1}} \leq 0$, $j \in K \setminus J_*$, and when $J_* \neq \emptyset$, each component of $\lambda^{J_*}(w, v)$ is positive. Furthermore,*

$$\begin{aligned} \mathcal{I}_K^*(w, v) &= \langle b_K(w, v) - \theta, v-w \rangle_{\Sigma^{-1}} \\ &= |A^{J_*} \theta|_{\Sigma^{-1}} |A^{J_*}(v-w)|_{\Sigma^{-1}} - \langle A^{J_*} \theta, A^{J_*}(v-w) \rangle_{\Sigma^{-1}}, \end{aligned}$$

and an optimal path is

$$x_K(t) = w + t b_K(w, v), \quad 0 \leq t \leq \alpha^{J_*}(w, v),$$

where

$$b_K(w, v) = \begin{cases} \alpha^{J_*}(w, v)(v-w) - R_{LJ_*} \alpha^{J_*}(w, v) B^{J_*}(v-w) + R_{LJ_*} B^{J_*} \theta, & \text{when } J_* \neq \emptyset, \\ \alpha^{J_*}(w, v)(v-w), & \text{when } J_* = \emptyset. \end{cases}$$

From Proposition 2, we can solve the VP (4.8) by identifying the set J_* for the pair of points v and w . One can find J_* by testing all subsets of K through the conditions from Proposition 2. In particular, when $|K| = 0$, Problem (4.8) only concerns paths in the interior of $\mathbb{R}_+^d$. We summarize the result in the following lemma, whose proof is the same as lemma 1 in El Kharroubi et al. (2012) and is, thus, omitted.

Lemma 5. *Given two points $v, w \in \mathbb{R}_+^d$ such that at least one of them is in the interior of $\mathbb{R}_+^d$, we have*

$$\mathcal{I}_\emptyset^*(w, v) = |\theta|_{\Sigma^{-1}} |v-w|_{\Sigma^{-1}} - \langle \theta, v-w \rangle_{\Sigma^{-1}},$$

and an optimal path is

$$x_\theta(t) = w + tb_\theta(w, v), \quad 0 \leq t \leq \frac{|v - w|_{\Sigma^{-1}}}{|\theta|_{\Sigma^{-1}}},$$

where $b_\theta(w, v) = \frac{|\theta|_{\Sigma^{-1}}}{|v - w|_{\Sigma^{-1}}}(v - w)$.

In the rest of section, we construct a subsolution to the VP (3.3). By Lemma 4, it suffices to find a function that satisfies the subsolution property with respect to (4.8). We start with $T(\cdot)$ being an appropriate linear function on $\mathbb{R}^d$, which is constructed later. We then rescale $T(\cdot)$ in an appropriate fashion by the functions $\mathcal{I}_K^*(\cdot, \cdot)$, $K \subset L$, and shift it to allow for the largest possible value at the origin.

Consider a nonempty subset $K \subset L$ and two points v, w satisfying Condition 1. We note that, from Proposition 2, if $w', v' \in \mathbb{R}_+^d$ also satisfy Condition 1, and $v' - w' = \alpha(v - w)$ for some $\alpha \in (0, \infty)$, then $\mathcal{I}_K^*(w', v') = \alpha \mathcal{I}_K^*(w, v)$. Therefore, it is proper to define $\mathcal{I}_K^*(v - w) = \mathcal{I}_K^*(w, v)$ for any $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1.

The following lemma plays a critical role in constructing a subsolution. Because R is a nonsingular $\mathcal{M}$ matrix, so is R^T (see Lemma 10 for properties of $\mathcal{M}$ matrices). The vector $u > 0$ in the lemma is chosen to satisfy $R^T u > 0$.

Lemma 6. *There exists a positive $d \times 1$ vector u , such that, given any $K \subset L$, and two points $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1 with respect to K , we have if $\mathcal{I}_K^*(w, v) = 0$, then $\langle v - w, u \rangle < 0$.*

Outline of Proof. We start with the observation that, if $\mathcal{I}_K^*(w, v) = 0$, then by Proposition 2, there exists an optimal path such that $\dot{\psi}(t) = \theta$ for all $t \in (0, \tau_\phi(v))$, where $\tau_\phi(v) = \inf \{t \geq 0 : \phi(t) = v\}$ given that $\phi(0) = w$. Then, $v - w = \theta \tau_\phi(v) + R\eta(\tau_\phi(v))$. Under Assumption 3, the vector θ can be expressed as a linear combination of the columns of R with negative coefficients, which enables us to eventually write $v - w$ as a linear combination of the columns of R with negative coefficients. The result follows from here. The detailed proof is provided in Section 8.3. $\square$

We now construct the linear function $T(\cdot)$ using the positive vector u obtained in Lemma 6. For $x \in \mathbb{R}^d$, we define

$$T(x) = \frac{\langle x, u \rangle}{u_1}, \quad (4.9)$$

where u_1 is the first element of u . Note that, for $x \in \mathbb{R}_+^d$, $T(x) \geq 0$.

For a subset $K \subset L$, define a direction set $\mathbb{D}_K \subset \mathbb{R}^d$ such that a vector $v \in \mathbb{D}_K$ if and only if v satisfies the following conditions:

1. $v_i = 0, \forall i \in K$
2. $|v| = 1$.
3. $\langle u, v \rangle \geq 0$, where u is as in Lemma 6.

Note that $\mathbb{D}_K$ is a closed set and, thus, compact because of the continuity of the norm $|\cdot|$. By the definition of $\mathcal{I}_K^*(v)$, it is easy to see that $\mathcal{I}_K^*(v)$ is continuous. We now show that $\mathcal{I}_K^*$ is bounded from below on the set $\mathbb{D}_K$.

Proposition 3. *There exists $\varepsilon > 0$ such that, for all $K \subset L$ and $v \in \mathbb{D}_K$, $\mathcal{I}_K^*(v) > \varepsilon$.*

Proof. It suffices to show that, for any $K \subset L$, there exists an $\varepsilon > 0$ such that, for any $v \in \mathbb{D}_K$, $\mathcal{I}_K^*(v) > \varepsilon$. From the extreme value theorem, $\mathcal{I}_K^*(v)$ attains the minimum in $\mathbb{D}_K$. It is also known that $\mathcal{I}_K^*(v) \neq 0$ for any $v \in \mathbb{D}_K$ by Lemma 6. Hence, there exists some positive number ε such that $\mathcal{I}_K^*(v) > \varepsilon$ for any $v \in \mathbb{D}_K$. $\square$

This proposition allows us to consider the maximum of the ratio T to $\mathcal{I}_K^*$ over the set $\mathbb{D}_K$. Let

$$r = \max \left\{ \frac{T(v)}{\mathcal{I}_K^*(v)} : \forall K \subsetneq L, \text{ and } \forall v \in \mathbb{D}_K \right\},$$

and define our subsolution to be

$$\bar{T}(v) = -\frac{T(v)}{r} + \inf_{x \in B} \frac{T(x)}{r}, \quad v \in \mathbb{R}_+^d. \quad (4.10)$$

Ideally, we would like to choose u such that the corresponding value of $\bar{T}(0)$ is as large as possible so that the growth rate of relative variance is small. Because of the complexity of the optimization problem, we could not solve it explicitly. In numerical studies, we tried different values of u , and this vector does not have a significant effect on algorithm performance.

Theorem 2. $\bar{T}$ is a subsolution to the VP (3.3).

Proof. From Lemma 4, it suffices to show that $\bar{T}(x) \leq 0$ for all $x \in B$, $\bar{T}$ is continuous and that, for all $w, v \in \mathbb{R}_+^d \setminus B$, which satisfy Condition 1 for some $K \subset L$, we have

$$\bar{T}(w) - \bar{T}(v) \leq \mathcal{I}_K^*(w, v).$$

Because the first two conditions are obviously satisfied, we only need to establish the third condition. First assuming $(v - w) \cdot u < 0$, then

$$\bar{T}(w) - \bar{T}(v) = \frac{T(v) - T(w)}{r} \leq 0 \leq \mathcal{I}_K^*(w, v).$$

Now if $(v - w) \cdot u \geq 0$, we have $(v - w)/|v - w| \in \mathbb{D}_K$, and

$$\begin{aligned} \bar{T}(w) - \bar{T}(v) &= \frac{T(v) - T(w)}{r} \\ &= \frac{|v - w|T((v - w)/|v - w|)}{r} \\ &\leq |v - w|\mathcal{I}_K^*((v - w)/|v - w|) \\ &= \mathcal{I}_K^*(v - w), \end{aligned}$$

where the second equality follows because T is a linear function, the third inequality follows from the definition of r , and the final equality follows from Proposition 2. $\square$

5. Splitting Algorithms for SRBMs

Particle-splitting and importance-sampling methods are the two most widely used methodologies to obtain numerical estimations of probabilities of rare events. Oftentimes, particle-splitting methods are used to simulate a class of rare events called first entrance probabilities. The main idea of particle splitting is to partition the state space of a process into a series of nested subsets. Then, the rare event of interest can be considered as a nested sequence of events. When a given subset is entered by a simulated particle for the first time, a number of offspring are generated at the entry point according to the assigned splitting mechanism. All the offspring follow the same law of the original process independently. More refined versions of splitting algorithms, such as RESTART, have been introduced (see Dean and Dupuis 2011 for details).

In Section 5.1, we construct the Euler discretization of the SRBM and show that the discretized SRBM and the SRBM are exponentially equivalent. In Sections 5.2 and 5.3, we use the solution/subsolution developed in Section 4 to develop algorithms to estimate the probability for our rare event of interest. Finally, in Section 5.4, we analyze the performance of our estimation algorithms.

5.1. Discrete Approximations for SRBMs

In this section, we construct a discrete approximation to the SRBM Z . Recall that the d -dimensional SRBM Z in $\mathbb{R}_+^d$ is given by

$$Z(t) = z + X(t) + \theta t + RY(t), \quad t \geq 0,$$

where X is a d -dimensional driftless Brownian motion with covariance matrix Σ and Y is the corresponding reflecting process. Fix $\Delta > 0$. Consider the discretized process X^Δ and $\tilde{X}^\Delta$ defined by $X^\Delta(t) = X((k-1)\Delta)$, and $\tilde{X}^\Delta(t) = X^\Delta(t) + \theta(k-1)\Delta$ for $t \in [(k-1)\Delta, k\Delta)$, and $k \in \mathbb{N}$. Define (Z^Δ, Y^Δ) as the solution of the Skorokhod problem for $\tilde{X}^\Delta$ with the initial value z . Then, we have

$$Z^\Delta(t) = z + \tilde{X}^\Delta(t) + RY^\Delta(t), \quad t \geq 0. \quad (5.1)$$

Introducing the large deviation scaling to Z^Δ , we define

$$Z_n^\Delta(t) = \frac{Z^\Delta(nt)}{n}, \quad t \geq 0.$$

Let $z_n \in \mathbb{R}_+^d \setminus (A_n \cup B)$ be the initial value of Z_n^Δ and $z_n \rightarrow 0$ as $n \rightarrow \infty$. The first entrance probability associated with Z_n^Δ is defined to be

$$p_n^\Delta(z_n) = \mathbb{P}(Z_n^\Delta(\tau_n^\Delta) \in B | Z_n^\Delta(0) = z_n),$$

where $\tau_n^\Delta = \inf\{t \geq 0 : Z_n^\Delta(t) \in A_n \cup B\}$. In the following, the first proposition establishes the exponential equivalence between Z_n and Z_n^Δ , and the second one characterizes the relative error between $p_n^\Delta(z_n)$ and $p_n(z_n)$.

Proposition 4. Under Assumption 1, if $\Delta \equiv \Delta(n) \rightarrow 0$ as $n \rightarrow \infty$, then for any $T \geq 0$ and $\varepsilon > 0$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \varepsilon \right) = -\infty.$$

Consequently, for $z_n \notin A_n \cup B$ satisfying $z_n \rightarrow 0$ as $n \rightarrow \infty$, under Assumptions 1 and 2,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log p_n^\Delta(z_n) = -\inf_{x \in B} I(x).$$

Outline of Proof. To estimate $\mathbb{P}(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \varepsilon)$, we use the Lipschitz continuity of the Skorokhod map and a standard estimate for the difference between a Brownian motion and its discretization. See Section 8.4 for the full proof. $\square$

Proposition 5. Let $\eta \in (0, 1/2)$ and $\Delta \equiv \Delta(n)$ satisfy $\lim_{n \rightarrow \infty} n\Delta^\eta = 0$, and assume that $z_n \notin A_n \cup B$ satisfying $\lim_{n \rightarrow \infty} z_n = 0$. Then, under Assumptions 1 and 2,

$$\left| \frac{\max_{x \in C_n} p_n^\Delta(x)}{\max_{x \in C_n} p_n(x)} - 1 \right| = O(n\Delta^\eta), \quad (5.2)$$

where $C_n = \{x \in \mathbb{R}_+^d : \sum_{k=1}^d x_k = \sum_{k=1}^d z_{n,k}\}$ with $z_{n,k}$, $k = 1, \dots, d$, being the k th component of z_n .

Ideally, Proposition 5 would be a statement of the form

$$\left| \frac{p_n^\Delta(z_n)}{p_n(z_n)} - 1 \right| = O(n\Delta^\eta). \quad (5.3)$$

Unfortunately, we were not able to prove such a statement. Because $p_n(x)$ has the same exponential decay rate for all $x \in C_n$, one can see that considering a maximum over the set C_n introduces at most a subexponential effect.

Outline of Proof of Proposition 5. Let $\eta \in (0, 1/2)$. We first establish the following lemma.

Lemma 7. If $\Delta(n) \rightarrow 0$ as $n \rightarrow \infty$, then for $\eta \in (0, 1/2)$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq \tau_n} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) = -\infty.$$

From this lemma, we can focus on the set on which the process Z_n^Δ deviates from Z_n within Δ^η distance over the time interval $[0, \tau_n]$. Now, take $\Delta \equiv \Delta(n)$ small enough such that $\varepsilon/n - \Delta^\eta > 0$ and $\varepsilon/n + \Delta^\eta < z_n$ and define

$$A_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \varepsilon/n - \Delta^\eta \right\}, \text{ and } B_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq 1 - \Delta^\eta \right\},$$

and

$$\tau_n^-(A_n^-) = \inf\{t \geq 0 : Z_n(t) \in A_n^-\}, \tau_n^-(B_n^-) = \inf\{t \geq 0 : Z_n(t) \in B_n^-\},$$

$$\tau_n^- = \inf\{t \geq 0 : Z_n(t) \in A_n^- \cup B_n^-\}.$$

Finally, for $\tilde{z} \in \mathbb{R}_+^d$, let $p_n^-(\tilde{z}) = \mathbb{P}(\tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z})$.

Lemma 8. For $\eta \in (0, 1/2)$, let $\Delta \equiv \Delta(n)$ satisfy $\lim_{n \rightarrow \infty} n\Delta^\eta = 0$. Then,

$$\frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \leq 1 + O(n\Delta^\eta),$$

where $C_n = \{\tilde{z} \in \mathbb{R}_+^d : \sum_{i=1}^d \tilde{z}_i = \sum_{i=1}^d z_{n,i}\}$.

Symmetrically, one can define $\tau_n^+ = \inf\{t \geq 0 : \sum_{k=1}^d Z_{n,k}(t) \geq 1 + \Delta^\eta, \text{ or } \sum_{k=1}^d Z_{n,k}(t) \leq \varepsilon/n + \Delta^\eta\}$, and $p^+(z_n) = \mathbb{P}(\sum_{k=1}^d Z_{n,k}(\tau_n^+) = 1 + \Delta^\eta | Z_n(0) = z_n)$. Similar to Lemma 8, it can be shown that

$$\frac{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \geq 1 + O(n\Delta^\eta). \quad (5.4)$$

Finally, to prove Proposition 5, we note that

$$\frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1 = \frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})} \frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1.$$

The first ratio in the right-hand side can be estimated using Lemma 7 such that $p_n^\Delta(z_n)/p_n^-(z_n) \leq 1 + o(e^{-Cn\Delta^{2\eta-1}})$ for some $C > 0$, and the second ratio has the estimate as in Lemma 8. Then, $\max_{x \in C_n} p_n^\Delta(x)/\max_{x \in C_n} p_n(x) \leq 1 + O(n\Delta^\eta)$. To show the other direction, one observes that

$$\frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1 = \frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})} \frac{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1.$$

The first ratio in the right-hand side can again be estimated using Lemma 7 such that $p_n^\Delta(z_n)/\max_{\tilde{z} \in C_n} p_n^+(\tilde{z}) \geq 1 + o(e^{-Cn\Delta^{2\eta-1}})$ for some $C > 0$, and the second ratio has the estimate as in (5.4). The result follows now. The proofs of the two lemmas and the complete proof of the proposition are provided in Section 8.5. $\square$

5.2. Particle-Splitting Methods

Following Dean and Dupuis (2009), we briefly describe the splitting algorithm to estimate the probabilities $\{p_n^\Delta(z_n)\}_{n \geq 1}$. Our state space $\mathbb{R}_+^d$ is partitioned according to a nested collection of sets $B = C_0^n \subset C_1^n \subset \dots \subset C_K^n \dots$. These sets are often constructed as the level sets of a particular function V , which is called the importance function in Dean and Dupuis (2009) and is defined in terms of the solution/subsolution of the VP (3.3).

Given the sequence of nested sets $\{C_j^n\}_{j \geq 0}$ and a splitting number m , the most simple splitting algorithm works in the following way. Define $l^n(x) = \min\{j \geq 0 : x \in C_j^n\}$. The algorithm generates a time-inhomogeneous branching process with generations indexed by $i \in \{0, 1, \dots, l^n(z_n)\}$.

- For generation 0, a single particle starts at z_n , and it evolves according to the law of Z_n^Δ with initial condition $Z_n^\Delta(0) = z_n$.
- Let N_{i-1}^n denote the number of particles in generation $i-1$, and for $j = 1, \dots, N_{i-1}^n$, let $X_{i-1,j}^n$ denote the position of the j th particles in generation $i-1$. The j th particle behaves according to the law of Z_n^Δ with initial condition $Z_n^\Delta(0) = X_{i-1,j}^n$, and it stops moving when reaching either the set A_n or the next level set $C_{l^n(X_{i-1,j}^n)-1}^n = C_{l^n(z_n)-i}^n$.
- If the j th particle in generation $i-1$ reaches A_n first, do nothing.
- If the j th particle in generation $i-1$ reaches $C_{l^n(z_n)-i}^n$ first, denoting by $X_{i,j}^n$ its location of entrance to $C_{l^n(z_n)-i}^n$, generate m number of new particles in generation i with position $X_{i,j}^n$.
- Once all the generations have been calculated, define the estimator

$$s_n = m^{-l^n(z_n)} N_{l^n(z_n)}^n. \quad (5.5)$$

Now, consider a subsolution $\bar{T}$. If the problem is two dimensional, we let

$$\bar{T}(v) = -I(v) + \inf_{x \in B} I(x), \quad (5.6)$$

where $I(v)$ is the explicit solution to the VP discussed in Section A.2. It is easy to check that this is indeed a subsolution, and $\bar{T}(0) = \inf_{x \in B} I(x)$. If the problem is three or higher dimensional, we use the subsolution constructed in (4.10), in which the linear function T on $\mathbb{R}^d$ is defined as in (4.9). Define the importance function $V(z) = \delta \bar{T}(z)/\log(m)$, where $\delta > 0$ is referred to as the level of the splitting algorithm. The level sets of the importance function V are defined to be

$$L_x = \{y \in \mathbb{R}_+^d : V(y) \leq x\}, \quad x \geq 0.$$

Now, define the sets C_j^n to be the level sets of V . More precisely, let

$$C_j^n = L_{(j-1)\delta/n}, \quad j \in \mathbb{N}.$$

In the following proposition, we assume that $z_n \notin A_n \cup B$ satisfying $\lim_{n \rightarrow \infty} z_n = 0$.

Proposition 6. *The splitting algorithm for Z^Δ is stable; that is, the total number of particles ever used grows subexponentially as $n \rightarrow \infty$. Letting $\{s_n^{(k)}\}_{k \in \mathbb{N}}$ be an independent and identically distributed sequence of copies of s_n , then, for each $N \in \mathbb{N}$,*

$$\mathbb{E} \left[\frac{1}{N} \sum_{k=1}^N s_n^{(k)} \right] = p_n^\Delta(z_n). \quad (5.7)$$

Furthermore, the variance of s_n can be measured using the following rate of decay of the second moment:

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}_{z_n} [s_n^2] = \inf_{x \in B} I(x) - \bar{T}(0). \quad (5.8)$$

Proposition 6 is a combination of proposition 7 and theorem 8 from Dean and Dupuis (2009), and its proof is presented in the appendix. It should be noted that Dean and Dupuis (2009) assume the rare event of interest is governed by a large deviations principle with a specific form that is different from what we consider in the current paper. However, the proofs of proposition 7 and theorem 8 do not depend on that assumption and still apply to the current setting.

5.3. RESTART splitting

The particle-splitting algorithm presented in Section 5.2 is very efficient when compared with standard Monte Carlo; however, there are some inefficiencies present. The source of inefficiency comes from the fact that most of the particles generated do not end up in the rare event set (in our case, set B) (Dean and Dupuis 2011). For those particles generated near set B by splitting, most of them still reach A_n before B , and it takes significant computational effort to simulate those trajectories.

Here, we briefly introduce the RESTART splitting algorithm that we implemented specifically for our problem. More details about the RESTART algorithm for rare event simulations can be found in Dean and Dupuis (2011). We first define a sequence of importance functions $\{V^n\}$ as follows.

$$\tilde{V}^n(y) = \Delta \left\lfloor \frac{nU(z_n) - nU(y)}{\Delta} \right\rfloor, \text{ and } V^n(y) = 0 \vee \tilde{V}^n(y),$$

where $\Delta = \log(r)$ with r being a fixed positive integer, z_n is the starting point, and $U(\cdot)$ is a subsolution satisfying the conditions in Definition 5. For each n , we define a sequence of nested sets $C_0^n \supset C_1^n \supset \dots \supset C_{j_n}^n \supset C_{j_n+1}^n$, where $C_0^n = \mathbb{R}_+^d$ and $C_{j_n+1}^n = \emptyset$ such that $y \in C_j^n \setminus C_{j+1}^n$ if and only if $V^n(y) = j\Delta$. For simplicity, we denote $C_j^n \setminus C_{j+1}^n$ by D_j .

The algorithm starts from an initial particle y_1 located at z_n . Each particle is assigned a killing threshold l when it is generated. The killing threshold for the initial particle is set to be zero. A particle is killed whenever it reaches any D_j such that $j < l$. A killed particle is not simulated further. During the simulation, whenever a particle moves from D_j to D_k , $k > j$, in one step, a total number of $m^{k-j} - 1$ (excluding the original particle) offspring are generated. For each integer $j < \alpha \leq k$, $(m - 1)m^{\alpha-j-1}$ of the offspring are assigned the killing threshold α . A particle is said to be stopped whenever it reaches set A_n or B . Notice that splitting can happen at the moment of a particle being stopped. In such case, all the offspring are also stopped. A stopped particle is not simulated further. The algorithm terminates when all the particles are stopped or killed.

Let $\mathbb{I}$ be the index set of those particles that reach set B . For a single trial of simulation, $p_n(z_n)$ is then approximated by $r_n = \sum_{i \in \mathbb{I}} \exp(V^n(y_i(\tau)))$, where $y_i(\tau)$ is the position of y_i when it is stopped. The following proposition summarizes the main results on the RESTART algorithm for Z_n^Δ .

Proposition 7. *The following hold for the RESTART algorithm:*

- i. *The estimator constructed from the RESTART algorithm is unbiased.*
- ii. *Suppose that $z_n \rightarrow 0$ with each $z_n \notin (A_n \cup B)$. Then,*

$$\liminf_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}_{z_n} [r_n^2] \geq \inf_{y \in B} \{I(y) + (\bar{T}(0) - \bar{T}(y)) \vee 0\}.$$

- iii. *Let w^n denote the sum of the lifetimes of all the particles simulated in a single trial. Then,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_{z_n} [w^n] = 0.$$

This proposition is a combination of results from Dean and Dupuis (2011), and its proof can be found in the appendix. The RESTART algorithm does not necessarily have a faster decay rate of its second moment than standard splitting. However, because of the lower computational effort per replication, it often has superior performance to standard splitting. In particular, there might be a slight increase in variance of the estimator because of the use of RESTART instead of splitting, but this increase is often made up for by not simulating particles that are very unlikely to hit the target set.

5.4. Estimation of Rare Event Probabilities

Our goal is to estimate the rare event probability $p_n(z_n)$, and we do this with the estimators s_n or r_n . To evaluate estimators, we introduce the following definition of efficiency.

Definition 6. An unbiased estimator α_n of $p_n(z_n)$ is said to have bounded relative error if $\mathbb{E}(\alpha_n^2) = O(p_n(z_n)^2)$, and it is weakly efficient or asymptotically optimal if, for each $\varepsilon > 0$, $\mathbb{E}(\alpha_n^2) = O(p_n(z_n)^{2-\varepsilon})$.

We focus our discussion on the estimator s_n derived in Section 5.2, but a similar analysis can be carried out for r_n that is considered in Section 5.3. A standard metric for the quality of the estimator is given by the following expected relative mean square error:

$$\begin{aligned} \text{rMSE}(s_n) &= \frac{1}{p_n(z_n)^2} \mathbb{E}[(p_n(z_n) - s_n)^2] \\ &= \underbrace{\frac{\mathbb{E}[(s_n - p_n^\Delta(z_n))^2]}{p_n(z_n)^2}}_{\text{relative variance}} + \underbrace{\frac{(p_n^\Delta(z_n) - p_n(z_n))^2}{p_n(z_n)^2}}_{\text{relative bias}}. \end{aligned} \quad (5.9)$$

In view of (5.9), the relative bias can be written as

$$\frac{(p_n^\Delta(z_n) - p_n(z_n))^2}{p_n(z_n)^2} = \left(\frac{p_n^\Delta(z_n)}{p_n(z_n)} - 1 \right)^2, \quad (5.10)$$

and using the fact that s_n is an unbiased estimator of $p_n^\Delta(z_n)$, the relative variance is equal to

$$\frac{\mathbb{E}[(s_n - p_n^\Delta(z_n))^2]}{p_n(z_n)^2} = \frac{\mathbb{E}[s_n^2]}{p_n(z_n)^2} - \left(\frac{p_n^\Delta(z_n)}{p_n(z_n)} \right)^2. \quad (5.11)$$

From Proposition 5 we expect that the relative bias (5.10) and the second term of the relative variance (5.11) grow at most at a subexponential rate. (Note that Proposition 5 is, unfortunately, not strong enough to provide precise control over this relative bias term, but we do know that, if it does grow, it does so in a subexponential fashion. It is our hope that a stronger statement of the form (5.3) can be achieved.) Next from Theorem 1 and Proposition 6,

$$\frac{\mathbb{E}[s_n^2]}{p_n(z_n)^2} = \exp \left[n \left(\inf_{x \in B} I(x) - \bar{T}(0) + o(1) \right) \right].$$

As mentioned earlier for the two-dimensional problem, we have that $\bar{T}(z) = -I(z) + \inf_{x \in B} I(x)$ and $\bar{T}(0) = \inf_{x \in B} I(x)$, and thus, the relative variance grows subexponentially in n . Consequently, for two-dimensional SRBMs, s_n is an asymptotically optimal estimator for $p_n^\Delta(z_n)$. For three and higher dimensions, recalling (4.9) and (4.10), we have that

$$\begin{aligned} \bar{T}(0) &= -\frac{T(0)}{r} + \frac{\inf_{x \in B} T(x)}{r} \\ &= \frac{\inf_{x \in B} \langle x, u \rangle}{\max \left\{ \frac{\langle v, u \rangle}{T_K^+(v)} : \forall K \subsetneq L, \text{ and } \forall v \in \mathbb{D}_K \right\}} \\ &= \frac{\min \{u_1, \dots, u_d\}}{\max \left\{ \frac{\langle v, u \rangle}{T_K^+(v)} : \forall K \subsetneq L, \text{ and } \forall v \in \mathbb{D}_K \right\}}. \end{aligned}$$

We observe that, because of the property of the vector u in Lemma 6, $\bar{T}(0) \in (0, \inf_{x \in B} I(x))$. Therefore, the relative variance term grows with exponential rate

$$\inf_{x \in B} I(x) - \bar{T}(0) < \inf_{x \in B} I(x).$$

As a point of comparison, suppose one were to use the standard Monte Carlo to estimate $p_n(z_n)$ and denote the estimator by $mc_n(z_n)$. Note that the bias terms would remain the same, but the relative variance term would grow with exponential rate of $\inf_{x \in B} I(x)$. Therefore, the estimator s_n has theoretically superior mean square error to $mc_n(z_n)$. Another point to mention is that the analysis of $\text{rMSE}(s_n)$ leaves out the computational work for generating

a single copy of s_n . However, Proposition 6 guarantees that the expected work per copy of s_n grows subexponentially, and thus, when normalized for expected work per replicate, s_n is still a superior estimator to $mc_n(z_n)$.

Finally, we summarize the efficiency of our estimators as follows:

- i. For two-dimensional SRBMs, both the relative bias and relative variance of our estimator grow at subexponential rates. Thus, the estimator is asymptotically optimal.
- ii. For three and higher dimensional SRBMs, the relative bias still grows subexponentially, and the relative variance grows with the exponential rate $\inf_{x \in B} I(x) - \bar{T}(0)$ that is strictly smaller than $\inf_{x \in B} I(x)$. Hence, our estimator is theoretically superior to the standard Monte Carlo estimator whose relative variance grows with exponential rate $\inf_{x \in B} I(x)$.
- iii. The splitting algorithm for Z_n^Δ is stable, which says the total number of particles generated grows subexponentially.

6. Numerical Examples

6.1. Splitting Algorithm

Before we run the splitting algorithm for a specific example, there are some initial steps needed to be done. We first obtain a subsolution $\bar{T}$ to the associated variation problem. If the problem is two dimensional, we let $T(v) = I(v)$, $\forall v \in \mathbb{R}_+^2$, where $I(v)$ is the explicit solution to the VP discussed in Section A.2. Then, we let $\bar{T}(v) = -T(v) + \inf_{x \in B} T(x)$. If the problem is three or higher dimensional, we can find the scaling factor r by solving the corresponding optimization problem with the help of any optimization solver and then construct the subsolution as defined in (4.10). Then, we choose a positive integer $m > 1$ as the number of new particles when a splitting occurs and choose $\delta \in (0, 1]$ as the level size. The importance function is then obtained as $V(z) = \delta \bar{T}(z) / \log(m)$. We then define a collection of level sets $\{C_j^n : j \in \mathbb{N}\}$, where $C_0^n = B$, $C_j^n = L_{(j-1)\delta/n}$, $j \in \mathbb{N}$, where $L_a = \{z : V(z) \leq a\}$. Finally, for a point x , its level can be calculated through the level function $l_n(x) = \min\{j \geq 0; x \in C_j^n\}$.

We conduct numerical experiments for one two-dimensional SRBM and one three-dimensional SRBM. The first splitting simulation (see Table 1) uses 1,000 runs with step size $1/(1,000 * n)$. We also show the level sets for the $n = 5$ case in Figure 1. The corresponding importance function can be found in the caption. The second splitting simulation (see Table 2) uses 200,000 runs with step size $1/(1,000 * n)$. Note that our step size does not comply with the requirement in Proposition 5 to achieve the asymptotic error bound. However, for every value of n used in the following simulations, $1/(1,000 * n) \leq 1/(n^{2.5})$, where the latter step size satisfies the requirement in Proposition 5. In this way, we increase the accuracy of simulation results when n is relatively small, and the polynomial factor is not too large as n increases. Simulations are run in Matlab with a computer having Intel i7-7500U CPU @ 2.70 GHz and RAM 8.00 GB (7.88 GB useable).

We also ran the standard Monte Carlo algorithm (see Table 3) for the three-dimensional example to compare with the results generated by the splitting algorithm. We ran standard Monte Carlo 1,000,000 times for $n = 5$, 10,000,000 times for $n = 10$, 100,000,000 times for $n = 15$, and 100,000,000 times for $n = 25$.

Table 1. Two-Dimensional SRBM Using the Splitting Algorithm

n	5	10	15
Estimate value	6.3000×10^{-4}	5.5500×10^{-6}	3.8000×10^{-8}
Time taken, seconds	6.4	386.8	871.3
Standard error	1.59×10^{-4}	6.63×10^{-7}	7.06×10^{-9}
95% confidence interval	$[3.18, 9.42] \times 10^{-4}$	$[4.25, 6.85] \times 10^{-6}$	$[2.42, 5.18] \times 10^{-8}$
Average number of particles	1.96	27.99	37.94
Standard number of particles	5.12	72.32	132.98
Maximum number of particles	55	496	1,288
n	20	30	40
Estimate value	2.5400×10^{-10}	1.0000×10^{-14}	3.5500×10^{-19}
Time taken, seconds	1,234.5	2,275.20	24,829.2
Standard error	6.24×10^{-11}	3.26×10^{-15}	8.80×10^{-20}
95% confidence interval	$[1.32, 3.76] \times 10^{-10}$	$[0.36, 1.64] \times 10^{-14}$	$[1.83, 5.27] \times 10^{-19}$
Average number of particles	36.85	40.32	340.04
Standard number of particles	185.37	304.73	1,282.36
Maximum number of particles	2,386	5,563	15,004

Note. Problem parameters: $z_n = (0.1/n, 0.1/n)$, $R = [1, 0; -1, 1]$, $\theta = [-2; 1]$, $\Sigma = [1, 0; 0, 1]$, $A_n = \{z \in \mathbb{R}_+^2 : \sum_{k=1}^2 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^2 : \sum_{k=1}^2 z_k \geq 1\}$.

Figure 1. Demonstration of Level Sets with $\tilde{T}(z) = -T(z) + 1$, Where $T(z)$ Is Equal to $3z_1 + z_2$ When $-2z_1 + z_2 \geq 0$ and Is Equal to $\sqrt{5(z_1^2 + z_2^2)} + 2z_1 - z_2$ Otherwise

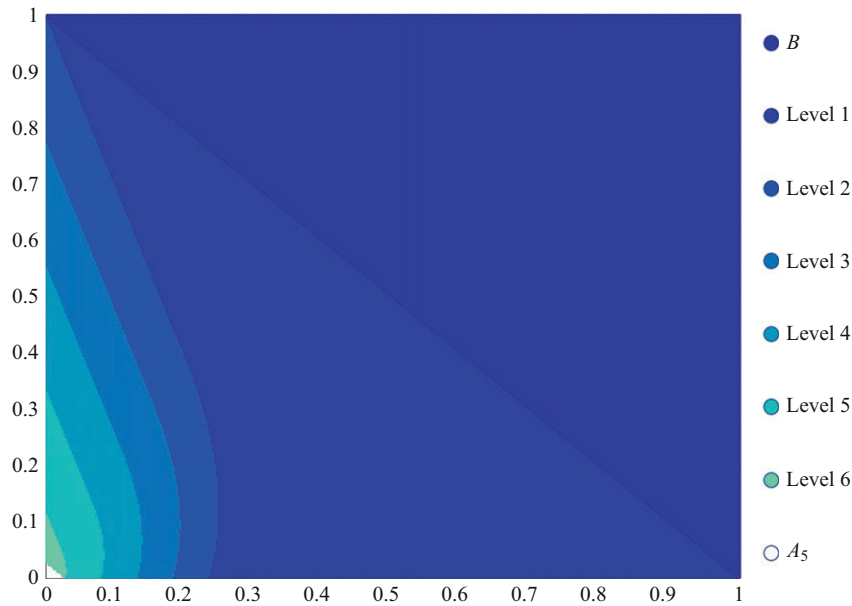


Table 2. Three-Dimensional SRBM Using the Splitting Algorithm

n	5	10	15
Estimate value	8.2030×10^{-3}	4.1910×10^{-4}	2.0565×10^{-5}
Time taken, seconds	159.8	537.9	986.8
Standard error	1.54×10^{-4}	1.11×10^{-5}	6.94×10^{-7}
95% confidence interval	$[7.90, 8.50] \times 10^{-3}$	$[3.97, 4.41] \times 10^{-4}$	$[1.92, 2.19] \times 10^{-5}$
Average number of particles	1.14	1.28	1.41
Standard number of particles	1.11	2.39	3.89
Maximum number of particles	10	55	136
n	20	30	40
Estimate value	1.0640×10^{-6}	2.1400×10^{-9}	5.0000×10^{-12}
Time taken, seconds	1,605.8	3,448.3	5,648.1
Standard error	4.24×10^{-8}	1.36×10^{-10}	5.52×10^{-13}
95% confidence interval	$[0.98, 1.15] \times 10^{-6}$	$[1.87, 2.41] \times 10^{-9}$	$[3.92, 6.08] \times 10^{-12}$
Average number of particles	1.55	1.82	2.08
Standard number of particles	5.59	9.39	13.66
Maximum number of particles	199	424	928

Note. Problem parameters: $z_n = (0.1/n, 0.1/n, 0.1/n)$, $R = [9, -4, -4; -3, 8, -4; -3, -4, 8]$, $\theta = [-2; -1; -1]$, $\Sigma = [2, 1, 1; 1, 2, 1; 1, 1, 3]$, $A_n = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \geq 1\}$.

Table 3. Three-Dimensional SRBM Using Standard Monte-Carlo

n	5	10	15	25
Estimate value	8.500×10^{-3}	4.261×10^{-4}	1.977×10^{-5}	7.000×10^{-8}
Time taken, seconds	340.4	3,556.4	57,293.8	70,311.8
Standard error	9.16×10^{-5}	6.53×10^{-6}	4.45×10^{-7}	2.65×10^{-8}
95% confidence interval	$[8.3, 8.6] \times 10^{-3}$	$[4.13, 4.39] \times 10^{-4}$	$[1.89, 2.06] \times 10^{-5}$	$[1.81, 12.19] \times 10^{-8}$

Note. Problem parameters: $z_n = (0.1/n, 0.1/n, 0.1/n)$, $R = [9, -4, -4; -3, 8, -4; -3, -4, 8]$, $\theta = [-2; -1; -1]$, $\Sigma = [2, 1, 1; 1, 2, 1; 1, 1, 3]$, $A_n = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \geq 1\}$.

Table 4. Two-Dimensional SRBM Using RESTART

n	5	10	15
Estimate value	8.853×10^{-4}	5.133×10^{-6}	3.796×10^{-8}
Time taken, seconds	435.8	492.4	660.7
Standard error	9.57×10^{-5}	9.24×10^{-7}	7.03×10^{-9}
95% confidence interval	$[6.977, 10.73] \times 10^{-4}$	$[3.322, 6.944] \times 10^{-6}$	$[2.419, 5.174] \times 10^{-8}$
Average number of particles	36.55	114.61	256.79
Standard number of particles	224.37	902.52	2.26×10^3
Maximum number of particles	4.671×10^3	2.675×10^4	5.949×10^4
n	20	30	40
Estimate value	2.611×10^{-10}	1.132×10^{-14}	3.146×10^{-19}
Time taken, seconds	852.4	965.0	1,107.0
Standard error	5.30×10^{-11}	3.19×10^{-15}	1.10×10^{-19}
95% confidence interval	$[1.573, 3.650] \times 10^{-10}$	$[0.507, 1.757] \times 10^{-14}$	$[0.992, 5.299] \times 10^{-19}$
Average number of particles	415.3	669.8	845.31
Standard number of particles	4.06×10^3	8.58×10^3	1.39×10^4
Maximum number of particles	1.114×10^5	3.267×10^5	7.602×10^5

Note. Problem parameters: $z_n = (0.1/n, 0.1/n)$, $R = [1, 0; -1, 1]$, $\theta = [-2; 1]$, $\Sigma = [1, 0; 0, 1]$, $A_n = \{z \in \mathbb{R}_+^2 : \sum_{k=1}^2 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^2 : \sum_{k=1}^2 z_k \geq 1\}$.

6.2. RESTART Algorithm

We applied the RESTART algorithm to simulate the same two- and three-dimensional examples as in the previous section. As expected, the RESTART algorithm required much less computational effort when n is large. Hence, we could run more trials and simulate the cases for a high value of n that would be very time-consuming when the splitting algorithm is applied.

Here, we give numerical results for a two- (see Table 4) and a three-dimensional SRBM (see Table 5). The simulation used 10,000 runs with step size $1/(1,000 * n)$. Simulations are run in Matlab with a computer having Intel i7-7500U CPU @ 2.70 GHz and RAM 8.00 GB (7.88 GB useable).

Table 5. Three-Dimensional SRBM Using RESTART

n	5	10	15
Estimate value	8.4640×10^{-3}	3.9923×10^{-4}	1.7487×10^{-5}
Time taken, seconds	486.0	530.5	543.3
Standard error	5.62×10^{-4}	3.83×10^{-5}	2.26×10^{-6}
95% confidence interval	$[7.3619, 9.5662] \times 10^{-3}$	$[3.2424, 4.7421] \times 10^{-4}$	$[1.3059, 2.1915] \times 10^{-5}$
Average number of particles	3.51	7.25	10.82
Standard number of particles	10.27	38.55	69.35
Maximum number of particles	207	914	1.923×10^3
n	20	30	40
Estimate value	9.3880×10^{-7}	2.2548×10^{-9}	4.7324×10^{-12}
Time taken, seconds	555.0	557.3	560.5
Standard error	1.40×10^{-7}	4.09×10^{-10}	1.05×10^{-12}
95% confidence interval	$[6.6418, 12.1341] \times 10^{-7}$	$[1.4530, 3.0567] \times 10^{-9}$	$[2.6744, 6.7904] \times 10^{-12}$
Average number of particles	14.7	20.2	25.85
Standard number of particles	115.55	196.67	278.54
Maximum number of particles	3.453×10^3	6.369×10^3	1.191×10^4

Note. Problem parameters: $z_n = (0.1/n, 0.1/n, 0.1/n)$, $R = [9, -4, -4; -3, 8, -4; -3, -4, 8]$, $\theta = [-2; -1; -1]$, $\Sigma = [2, 1, 1; 1, 2, 1; 1, 1, 3]$, $A_n = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \geq 1\}$.

6.3. Algorithm Comparison

Following Asmussen and Glynn (2007), we report the value obtained from multiplying sample variance by the CPU time (see Table 6) for each of the three algorithms discussed. We denote such value by sample variance per unit of CPU time (SVar/CPU). SVar/CPU provides a fair comparison between standard Monte Carlo, the splitting algorithm, and the RESTART algorithm. In general, an algorithm with smaller SVar/CPU is preferred. As seen in Table 6, splitting and RESTART always have superior performance to standard Monte Carlo.

Table 6. Comparison of Computational Cost Between Splitting, RESTART, and Standard Monte-Carlo

n	5	10	15
SVar/CPU for standard Monte Carlo	2.8578	1.5147	1.1327
SVar/CPU for splitting algorithm	0.7555	0.0131	9.5075×10^{-5}
SVar/CPU for RESTART algorithm	1.5368	0.0078	2.7730×10^{-5}
n	25	30	40
SVar/CPU for standard Monte Carlo	0.0049	N/A	N/A
SVar/CPU for splitting algorithm	2.9685×10^{-9}	1.2847×10^{-11}	3.4440×10^{-16}
SVar/CPU for RESTART algorithm	2.9927×10^{-10}	9.3274×10^{-13}	6.1798×10^{-18}

Note. Problem parameters: $z_n = (0.1/n, 0.1/n, 0.1/n)$, $R = [9, -4, -4; -3, 8, -4; -3, -4, 8]$, $\theta = [-2; -1; -1]$, $\Sigma = [2, 1, 1; 1, 2, 1; 1, 1, 3]$, $A_n = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \leq 0.15/n\}$, $B = \{z \in \mathbb{R}_+^3 : \sum_{k=1}^3 z_k \geq 1\}$.

7. Discussion

In this project, we develop particle-based simulation algorithms for estimating rare event probabilities related to SRBMs in a nonnegative orthant. We find that splitting-type algorithms provide much better performance than the standard Monte Carlo method. Furthermore, we find that the RESTART splitting algorithm is superior to the regular splitting algorithm in terms of operating time. Following Dean and Dupuis (2009, 2011), our algorithms are based on subsolutions to the variational problems associated with our rare events. In particular, the work Avram et al. (2001) develops solutions to these variational problems in two dimensions. For three and higher dimensions, we construct new subsolutions to the variational problem. Because of the results of Dean and Dupuis (2009, 2011), we are able to show theoretically that our splitting algorithm has superior performance to standard Monte Carlo in two and higher dimensions. This theoretical result is verified in numerical simulations in which we see that our algorithm performs significantly better than standard Monte Carlo (see Table 6).

Future directions for this work include developing subsolutions in three and higher dimensions that are closer to the solution and, thus, enable simulation algorithms with better performance. Another possible direction is to consider more general reflecting diffusions, that is, reflecting diffusions with state-dependent drift and variance.

8. Proofs

We provide the complete proofs of Propositions 1, 4, and 5; Theorem 1, and Lemma 6 in this section. At the beginning of each section, we restate the statement to be proved. For ease of notation, we let $D = \Sigma^{-1}$.

8.1. Proof of Proposition 1

Proposition 1. Assume $z_n \rightarrow z \in \mathbb{R}_+^d$. Under Assumptions 1 and 2, there exists $c \in (0, \infty)$ such that

$$\limsup_{n \rightarrow \infty} \sup_{z \in \mathbb{R}_+^d} \frac{1}{n} \log \mathbb{E}_{z_n} (e^{cn\tau_n}) < \infty.$$

Here, $\mathbb{E}_{z_n}$ denotes the expectation conditioning on $Z_n(0) = z_n$.

To show Proposition 1, we introduce the following Lyapunov function introduced in Dupuis and Williams (1994). Recall that R_i is the i th column of the reflection matrix R ; for a $d \times d$ matrix M , its trace is denoted as $\text{tr}(M)$, and for $f \in \mathcal{AC}([0, \infty); S)$, $\dot{f}$ and $\ddot{f}$ denote its gradient vector and Hessian matrix.

Theorem 3 (Dupuis and Williams 1994). Under Assumption 2, and the assumption that R is completely S , there exists a Lyapunov function $L : \mathbb{R}_+^K \rightarrow \mathbb{R}_+$ such that

- $L \in C^2(\mathbb{R}_+^d \setminus \{0\})$.
- Given $M_0 > 0$, there exists an $M_1 > 0$ such that, when $|z| \geq M_1$, we have $L(z) \geq M_0$.
- Given $\varepsilon > 0$, there exists $M_2 > 0$ such that, when $|z| \geq M_2$, we have $|\ddot{L}(z)| \leq \varepsilon$.
- There exists $c_0 > 0$ such that

$$\langle \dot{L}(z), \theta \rangle \leq -c_0, \text{ for all } z \in \mathbb{R}_+^d.$$

$$\langle \dot{L}(z), R_i \rangle \leq -c_0, \text{ for all } z \in \partial \mathbb{R}_+^d \setminus \{0\}.$$

- $L(\alpha z) = \alpha L(z)$ for all $\alpha \geq 0$ and $z \in \mathbb{R}_+^d$.

Some consequences of these properties are obtained in Budhiraja and Lee (2007), which are listed as follows.

- vi. For every $M_3 > 0$, there exists a $\gamma(M_3) > 0$ such that $\sup_{|z| \leq M_3} L(z) \leq \gamma(M_3)$.
- vii. There exists an $M_4 > 0$ such that $\sup_{z \in \mathbb{R}_+^d \setminus \{0\}} |\dot{L}(z)| \leq M_4$.
- viii. There exists $0 < c_1 \leq c_2 < \infty$ such that $c_1|z| \leq L(z) \leq c_2|z|$.

Lemma 9 (Budhiraja and Lee 2007). Fix $x \in \mathbb{R}_+^d$ and $\Delta > 0$. For $m \in \mathbb{N}$, define

$$v_m = \sup_{(m-1)\Delta \leq t \leq m\Delta} \left| \int_{(m-1)\Delta}^t \langle \dot{L}(Z(s)), dX(s) \rangle \right|.$$

Then, for any $\kappa > 0$ and $m, n \in \mathbb{N}, m \leq n$, we have

$$\mathbb{E}_x \left(\exp \left\{ \kappa \sum_{i=m}^n v_i \right\} \right) \leq \left(2\sqrt{2} \exp \left\{ \kappa^2 M_4^2 \gamma_0 \Delta \right\} \right)^{n-m+1},$$

where M_4 is as in Theorem 3(vii) and γ_0 is a positive constant only depending on the norm of Σ^{-1} .

Proof of Proposition 1. Define $\tau_\epsilon = \inf \{t \geq 0 : \sum_{k=1}^d Z_k(t) \leq \epsilon\}$ and then $n\tau_n \leq \tau_\epsilon$. Thus, it suffices to show that there exists $c > 0$ such that

$$\limsup_{n \rightarrow \infty} \sup_{z \in \mathbb{R}_+^d \setminus B} \frac{1}{n} \log \mathbb{E}_z(e^{c\tau_\epsilon}) < \infty.$$

The proof idea is adapted from that of theorem 4.3 of Budhiraja and Lee (2007). From Theorem 3(iii), choose $M_2 > 0$ such that, when $\sum_{k=1}^d z_k \geq M_2$, $\text{tr}(\ddot{L}(z)\Sigma) < c_0$, where c_0 is as in Theorem 3(iv). Let $M_0 > M_2$, and define $\kappa_\epsilon = M_0/\epsilon$. For $N \in \mathbb{N}$, consider the set $A_N = \{\omega \in \Omega : \inf_{0 \leq s \leq N\Delta} \sum_{k=1}^d Z_k(s) > \epsilon\}$. For $\omega \in A_N$ and $m \leq N$, from Ito's formula, we have

$$\begin{aligned} L(Z(m\Delta)) &= L(Z((m-1)\Delta)) + \int_{(m-1)\Delta}^{m\Delta} \frac{1}{2} \text{tr}(\ddot{L}(Z(s))\Sigma) ds \\ &\quad + \int_{(m-1)\Delta}^{m\Delta} \langle \dot{L}(Z(s)), \theta \rangle ds + \int_{(m-1)\Delta}^{m\Delta} \langle \dot{L}(Z(s)), dX(s) \rangle \\ &\quad + \sum_{i=1}^d \int_{(m-1)\Delta}^{m\Delta} \langle \dot{L}(Z(s)), R_i \rangle dY_i(s). \end{aligned}$$

Multiplying both sides of this equation by κ_ϵ , from Theorem 3(iv), (v), and (viii), we have, for $m \leq n$ on A_n ,

$$c_1 M_0 < \kappa_\epsilon L(Z(m\Delta)) \leq \kappa_\epsilon L(Z((m-1)\Delta)) - \frac{c_0 \Delta}{2} + \kappa_\epsilon v_m,$$

which implies that

$$c_1 M_0 < \kappa_\epsilon L(Z(N\Delta)) \leq \kappa_\epsilon L(nz_n) - \frac{c_0 N\Delta}{2} + \kappa_\epsilon \sum_{m=1}^N v_m. \quad (8.1)$$

From (8.1) and using Lemma 9, we have, for any $\kappa > 0$,

$$\begin{aligned} \mathbb{P}_z(A_N) &\leq \mathbb{P}_z \left(\kappa_\epsilon \sum_{m=1}^N v_m \geq c_1 M_0 + \frac{c_0 N\Delta}{2} - \kappa_\epsilon L(nz_n) \right) \\ &= \mathbb{P}_z \left(\kappa \kappa_\epsilon \sum_{m=1}^N v_m \geq \kappa \left(c_1 M_0 + \frac{c_0 N\Delta}{2} - \kappa_\epsilon L(nz_n) \right) \right) \\ &\leq \exp \left\{ -\kappa \left(c_1 M_0 + \frac{c_0 N\Delta}{2} - \kappa_\epsilon L(nz_n) \right) \right\} \mathbb{E}_x \left[\exp \left\{ \kappa \kappa_\epsilon \sum_{m=1}^N v_m \right\} \right] \\ &\leq \exp \{ \kappa \kappa_\epsilon L(nz_n) \} \exp \left\{ N\Delta [\log(2\sqrt{2})/\Delta + \kappa^2 \kappa_\epsilon^2 M_4^2 \gamma_0 - \kappa c_0/2] - \kappa c_1 M_0 \right\}. \end{aligned}$$

We can choose Δ large enough and κ small enough such that $-\eta \equiv \log(2\sqrt{2})/\Delta + \kappa^2 \kappa_\epsilon^2 M_4^2 \gamma_0 - \kappa c_0/2 < 0$. Therefore, we have

$$\mathbb{P}_z(A_n) \leq \exp \{ \kappa \kappa_\epsilon L(nz_n) \} \exp \{ -\eta N\Delta \}.$$

For $t > 0$, there exists $n_0 \in \mathbb{N}$ such that $t \in [n_0\Delta, (n_0 + 1)\Delta]$, and we have

$$\mathbb{P}_{nz_n}(\tau_\epsilon > t) \leq \mathbb{P}_{nz_n}(A_{n_0}) \leq \exp\{\kappa\kappa_\epsilon L(nz_n)\} \exp\{-\eta n_0\Delta\} \leq \exp\{\kappa\kappa_\epsilon L(nz_n) + \eta\Delta\} \exp\{-\eta t\}.$$

When $z \in \mathbb{R}_+^d$ satisfies that $\sum_{k=1}^d z_k = 0$ or $\sum_{k=1}^d z_k \geq 1$, we have $\tau_n = 0$ when n is large enough. When $0 < \sum_{k=1}^d z_k < 1$, we have, for $c \in (0, \eta)$,

$$\begin{aligned} \mathbb{E}_{z_n}(e^{cn\tau_n}) &\leq \mathbb{E}_{nz_n}(e^{c\tau_\epsilon}) = 1 + \int_0^\infty ce^{ct} \mathbb{P}_{nz_n}(\tau_\epsilon > t) dt \\ &\leq 1 + \frac{c}{\eta - c} \exp\{\kappa\kappa_\epsilon L(nz_n) + \eta\Delta\} \\ &\leq 1 + \frac{c}{\eta - c} \exp\{\kappa\kappa_\epsilon c_2 |nz_n| + \eta\Delta\}. \end{aligned}$$

The result follows.

8.2. Proof of Theorem 1

Theorem 1. Assume $z_n \in \mathbb{R}_+^d / (A_n \cup B)$ for each n , and $z_n \rightarrow 0$. Under Assumptions 1 and 2,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) = -\inf_{x \in B} I(x).$$

Proof. We first show that

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \leq -\inf_{x \in B} I(x).$$

From Proposition 1, it follows that

$$\lim_{t \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(\tau_n > t) \leq \lim_{t \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \frac{\mathbb{E}_{z_n}(e^{cn\tau_n})}{e^{cnt}} = -\infty,$$

where $\mathbb{P}_{z_n}$ and $\mathbb{E}_{z_n}$ are the probability and expectation conditioning on $Z_n(0) = z_n$. Therefore, we can take $T > 0$ such that

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(\tau_n > T) < -\inf_{x \in B} I(x).$$

The following proof is similar to that of lemma 5.7.21 in Dembo and Zeitouni (2010). Define the following closed set of $\mathcal{C}([0, T]; \mathbb{R}_+^d)$:

$$F = \{\phi \in \mathcal{C}([0, T]; \mathbb{R}_+^d) : \psi \in \mathcal{C}([0, T]; \mathbb{R}^d), \phi(t) = \Gamma(\psi)(t) \in B \text{ for some } t \in [0, T]\}.$$

Then,

$$\begin{aligned} p_n(z_n) &= \mathbb{P}_{z_n}(Z_n(\tau_n) \in B, \tau_n > T) + \mathbb{P}_{z_n}(Z_n(\tau_n) \in B, \tau_n \leq T) \\ &\leq \mathbb{P}_{z_n}(\tau_n > T) + \mathbb{P}_{z_n}(Z_n \in F). \end{aligned}$$

Also note that, for $T > 0$, we have

$$\inf_{x \in B} I(x) \leq \inf_{\phi \in F} I_{T,0}(\phi) < \infty.$$

According to lemma 1.2.16 in Dembo and Zeitouni (2010),

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) &\leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log (\mathbb{P}_{z_n}(\tau_n > T) + \mathbb{P}_{z_n}(Z_n \in F)) \\ &\leq \max \left[\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(\tau_n > T), \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n}(Z_n \in F) \right] \\ &\leq -\inf_{\phi \in F} I_{T,0}(\phi) \leq -\inf_{x \in B} I(x). \end{aligned}$$

Now, we show the lower bound

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \geq - \inf_{x \in B} I(x).$$

Note that

$$\inf_{x \in B} I(x) \geq \inf_{\psi \in \mathcal{AC}_0([0, \infty), \mathbb{R}^d), \tau_\psi < \infty} \frac{1}{2} \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds,$$

where $\tau_\psi = \inf \{t \geq 0 : \Gamma(\psi)(t) \in B\}$. Thus, it suffices to establish that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \geq - \frac{1}{2} \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds \quad (8.2)$$

for any $\psi \in \mathcal{AC}([0, \infty), \mathbb{R}^d)$ such that $\psi(0) = z$ and $\tau_\psi < \infty$. We claim that the optimal solutions of the VP (3.3) would never return to the origin before reaching B . Indeed, let $t_0 \geq 0$ denote the last time an optimal solution ϕ visits the origin before reaching B and (ψ, ϕ, η) be a corresponding optimal triple. Define $\tilde{\psi}(t) = \psi(t + t_0) - \psi(t_0)$, and then from Budhiraja and Dupuis (1999), $\Gamma(\tilde{\psi})(t) = \phi(t + t_0)$ with $\Gamma(\tilde{\psi})(0) = 0$ and $\tau_{\tilde{\psi}} = \tau_\psi - t_0$. Thus,

$$\frac{1}{2} \int_0^{\tau_{\tilde{\psi}}} |\dot{\tilde{\psi}}(s) - \theta|_D^2 ds = \frac{1}{2} \int_{t_0}^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds \leq \frac{1}{2} \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds. \quad (8.3)$$

Therefore, we must have $\psi(t) = \theta t$ for $t \in [0, t_0]$ because ϕ is an optimal solution and $\psi(0) = 0$. Noting that $\phi(0) = \phi(t_0) = 0$ and each component of θt is monotone, we must have that $t_0 = 0$, which says the optimal path ϕ will not return to the origin once leaving it. We only need to consider ψ such that $\sum_{k=1}^d \Gamma(\psi)_k(s) > 0$ for all $s \in (0, \tau_\psi]$. Fix such a ψ . Let $\varphi^z(t) = \tilde{z} + \psi(t)$ with $\sum_{k=1}^d \tilde{z}_k > 0$. It is clear that

$$\int_0^{\tau_\psi} |\dot{\varphi}^z(s) - \theta|_{\Sigma^{-1}}^2 ds = \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds.$$

Fix $\kappa > 0$, and let $B_\kappa = \{\tilde{z} \in \mathbb{R}_+^d : \sum_{k=1}^d \tilde{z}_k = \kappa\}$. For $\kappa, \delta > 0$, define the following open set:

$$G_\kappa(\varphi; \delta) = \bigcup_{\tilde{z} \in B_\kappa} \left\{ \eta \in \mathcal{C}([0, \tau_\psi]; \mathbb{R}_+^d) : -\delta < (1 - \delta) \sum_{k=1}^d \eta_k(t) - \sum_{k=1}^d \Gamma(\varphi^z)_k(t) < \delta \text{ for all } 0 \leq t \leq \tau_\psi \right\}.$$

For $\kappa \in (0, 1)$, there exists $\delta_0 > 0$ and $N \in \mathbb{N}$ such that, if $0 < \delta \leq \delta_0$, then for all $\eta \in G_\kappa(\varphi; \delta)$,

$$\sum_{i=1}^d \eta_i(t) > \left[\inf_{0 \leq t \leq \tau_\psi} \sum_{i=1}^d \Gamma(\varphi^z)_i(t) - \delta \right] / (1 - \delta) > \epsilon / N,$$

for all $t \in [0, \tau_\psi]$, and

$$\sum_{i=1}^d \eta_i(\tau_\psi) > \left[\sum_{i=1}^d \Gamma(\varphi^z)_i(\tau_\psi) - \delta \right] / (1 - \delta) \geq 1.$$

Based on these two displays, we conclude that G_κ is an open set of sample paths such that

$$\inf_{Z_n(0) \in B_\kappa} \mathbb{P}(Z_n \in G_\kappa(\varphi; \delta)) \leq \inf_{Z_n(0) \in B_\kappa} \mathbb{P}(Z_n \text{ hits } B \text{ before hitting } A_n).$$

Assume that n is large enough such that $\kappa > \sum_k z_{n,k} > \epsilon/n$. From the Markov property of Z_n , we have that

$$p_n(z_n) \geq \mathbb{P}_{z_n}(Z_n \text{ hits } B_\kappa \text{ before hitting } A_n) \mathbb{P}(Z_n \text{ hits } B \text{ before hitting } A_n | Z_n(0) \in B_\kappa).$$

Using the minimality property of the one-dimensional Skorokhod map, we see that $\sum_{k=1}^d Z_{n,k}(t) \geq \Gamma_1(\sum_{k=1}^d z_{n,k} + \sum_{k=1}^d \tilde{X}_{n,k})(t)$ for all $t \geq 0$, where $\tilde{X}_n(t) = \theta t + X(nt)/n$ and Γ_1 is the one-dimensional Skorokhod map. Thus, we have

$$\begin{aligned} & \mathbb{P}_{z_n}(Z_n \text{ hits } B_\kappa \text{ before hitting } A_n) \\ & \geq \mathbb{P}_{\sum_{k=1}^d z_{n,k}} \left(\Gamma_1 \left(\sum_{k=1}^d z_{n,k} + \sum_{k=1}^d \tilde{X}_k \right) \text{ reaches } \kappa \text{ before reaching } \epsilon/n \right) \end{aligned}$$

$$\begin{aligned}
&= \mathbb{P}_{\sum_{k=1}^d z_{n,k}} \left(\sum_{k=1}^d z_{n,k} + \sum_{k=1}^d \tilde{X}_k \text{ reaches } \kappa \text{ before reaching } \epsilon/n \right) \\
&= \frac{1 - e^{-n\gamma b_n}}{e^{n\gamma a_n} - e^{-n\gamma b_n}},
\end{aligned}$$

where $\gamma = \frac{2|\sum \theta|}{1 \cdot \sum 1}$, $a_n = \kappa - \sum_{k=1}^d z_{n,k}$, and $b_n = \sum_{k=1}^d z_{n,k} - \frac{\epsilon}{n}$. Thus,

$$\begin{aligned}
&\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}_{z_n} (Z_n \text{ hits } B_\kappa \text{ before hitting } A_n) \\
&\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \frac{1 - e^{-n\gamma b_n}}{e^{n\gamma a_n} - e^{-n\gamma b_n}} \\
&\geq \liminf_{n \rightarrow \infty} \left(\frac{\log(1 - e^{-n\gamma b_n})}{n} - \gamma a_n \right) \\
&= -\gamma \kappa.
\end{aligned}$$

We next note that

$$\begin{aligned}
&\liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(Z_n \text{ hits } B \text{ before hitting } A_n | Z_n(0) \in B_\kappa) \\
&\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \inf_{Z_n(0) \in B_\kappa} \mathbb{P}(Z_n \text{ hits } B \text{ before hitting } A_n) \\
&\geq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \inf_{Z_n(0) \in B_\kappa} \mathbb{P}(Z_n \in G_\kappa(\varphi; \delta)) \\
&\geq -\sup_{x \in B_\kappa} \inf_{\eta \in G_\kappa(\varphi; \delta)} I_{\tau_\psi, x}(\eta) \\
&\geq -\sup_{x \in B_\kappa} \frac{1}{2} \int_0^{\tau_\psi} \left| \frac{1}{1-\delta} \dot{\varphi}^x(s) - \theta \right|_{\Sigma^{-1}}^2 ds \\
&= -\frac{1}{2} \int_0^{\tau_\psi} \left| \frac{1}{1-\delta} \dot{\psi}(s) - \theta \right|_{\Sigma^{-1}}^2 ds. \tag{8.4}
\end{aligned}$$

Letting $\delta \downarrow 0$, we conclude that

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \geq -\gamma \kappa - \frac{1}{2} \int_0^{\tau_\psi} |\dot{\psi}(s) - \theta|_{\Sigma^{-1}}^2 ds.$$

Finally, letting $\kappa \downarrow 0$, (8.2) follows. $\square$

8.3. Proof of Lemma 6

Lemma 6. *There exists a positive $d \times 1$ vector u such that, given any $K \subset L$, and two points $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1 with respect to K , we have, if $\mathcal{I}_K^*(w, v) = 0$, then $\langle v - w, u \rangle < 0$.*

To prove Lemma 6, we list some properties of nonsingular $\mathcal{M}$ -matrices.

Lemma 10 (Plemmons 1977). *If R is a nonsingular $\mathcal{M}$ -matrix, then*

- i. $R^{-1} \geq 0$.
- ii. *The diagonal elements of R are positive, and off-diagonal elements are nonpositive.*
- iii. R^T is a nonsingular $\mathcal{M}$ -matrix.
- iv. *There exists $x > 0$ with $Rx > 0$.*
- v. *There exists a positive diagonal matrix Λ such that $R\Lambda$ has all positive row sums.*

Proof of Lemma 6. If $\mathcal{I}_K^*(w, v) = 0$, then, by Proposition 2, there exists an optimal path such that $\dot{\psi}(t) = \theta$ for all $t \in (0, \tau_\phi(v))$, where $\tau_\phi(v) = \inf \{t \geq 0 : \phi(t) = v\}$ given that $\phi(0) = w$. Hence,

$$\begin{aligned}
v - w &= \phi(\tau_\phi(v)) - \phi(0) = w + \theta \tau_\phi(v) + R\eta(\tau_\phi(v)) - w \\
&= \theta \tau_\phi(v) + R\eta(\tau_\phi(v)), \tag{8.5}
\end{aligned}$$

where $\eta_i(\tau_\phi(v)) = 0$, $\forall i \in L \setminus K$ because $\phi(t) \in F_K$ for $t \in (0, \tau_\phi(v))$. From Assumption 3 and the invertibility of reflection matrix R , we can write θ as $\theta = \alpha_1 R_1 + \alpha_2 R_2 + \dots + \alpha_d R_d$, where $\alpha_i < 0$, $\forall i$, and R_i is the i th column of R . By Lemma 10(v), we can further assume that R has all positive row sums. We then rewrite (8.5) as

$$\begin{aligned} v - w &= \theta \tau_\phi(v) + R \eta(\tau_\phi(v)) \\ &= \tau_\phi(v) [\alpha_1 R_1 + \alpha_2 R_2 + \dots + \alpha_d R_d] + [\eta_1(\tau_\phi(v)) R_1 + \eta_2(\tau_\phi(v)) R_2 + \dots + \eta_d(\tau_\phi(v)) R_d] \\ &= (\tau_\phi(v) \alpha_1 + \eta_1(\tau_\phi(v))) R_1 + \dots + (\tau_\phi(v) \alpha_d + \eta_d(\tau_\phi(v))) R_d. \end{aligned}$$

Without loss of generality, suppose $K = \{1, 2, \dots, m\}$, where $m < d$. Then, the first m entries of $v - w = [(\tau_\phi(v) \alpha_1 + \eta_1(\tau_\phi(v))) R_1 + \dots + (\tau_\phi(v) \alpha_d + \eta_d(\tau_\phi(v))) R_d]$ are all zeros. We show that $\tau_\phi(v) \alpha_i + \eta_i(\tau_\phi(v)) \leq 0$, $\forall i \in K$. Suppose that there exists some $k_1, \dots, k_n \in K$ such that $\tau_\phi(v) \alpha_{k_i} + \eta_{k_i}(\tau_\phi(v)) > 0$ for $i \in \{1, 2, \dots, n\}$. Let $\tilde{k} = \arg \max_{k \in \{k_1, \dots, k_n\}} \{\tau_\phi(v) \alpha_k + \eta_k(\tau_\phi(v))\}$. Then, by the assumption that R has positive row sums and Lemma 10(ii),

$$(\tau_\phi(v) \alpha_1 + \eta_1(\tau_\phi(v))) R_{\tilde{k}1} + \dots + (\tau_\phi(v) \alpha_d + \eta_d(\tau_\phi(v))) R_{\tilde{k}d} > 0,$$

which contradicts the fact that the first m entries of $v - w = [(\tau_\phi(v) \alpha_1 + \eta_1(\tau_\phi(v))) R_1 + \dots + (\tau_\phi(v) \alpha_d + \eta_d(\tau_\phi(v))) R_d]$ are all zeros.

Hence, $\tau_\phi(v) \alpha_i + \eta_i(\tau_\phi(v)) \leq 0$, $\forall i \in K$. Because $\eta_i(\tau_\phi(v)) = 0$, $\forall i \in L \setminus K$ and $\alpha_i < 0$, $\forall i$, we also have $\tau_\phi(v) \alpha_i + \eta_i(\tau_\phi(v)) < 0$, $\forall i \in L \setminus K$. If we let ξ be a $d \times 1$ vector such that $\xi_i = \tau_\phi(v) \alpha_i + \eta_i(\tau_\phi(v))$, then $\xi \leq 0$ and $\xi \neq 0$. By Lemma 10(iii) and (iv), there exists a positive vector u such that $R^T u > 0$. Therefore,

$$\begin{aligned} (v - w) \cdot u &= [(\tau_\phi(v) \alpha_1 + \eta_1(\tau_\phi(v))) R_1 + \dots + (\tau_\phi(v) \alpha_d + \eta_d(\tau_\phi(v))) R_d] \cdot u \\ &= u^T R \xi < 0. \quad \square \end{aligned}$$

8.4. Proof of Proposition 4

Proposition 4. Under Assumption 3, if $\Delta \equiv \Delta(n) \rightarrow 0$ as $n \rightarrow \infty$, then for any $T \geq 0$ and $\varepsilon > 0$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \varepsilon \right) = -\infty.$$

Consequently, for $z_n = z/n \notin A_n \cup B$ satisfying $z_n \rightarrow 0$ as $n \rightarrow \infty$, under Assumptions 1 and 2,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log p_n^\Delta(z_n) = -\inf_{x \in B} I(x).$$

Proof. Let W be a standard Brownian motion and, for a fixed $\Delta > 0$, define its discretization $W^\Delta(t) = W((k-1)\Delta)$ for $t \in [(k-1)\Delta, k\Delta)$ and $k \in \mathbb{N}$. Let $T > 0$, and consider $\mathbf{W}_T = \{W(t) : 0 \leq t \leq T\}$ and $\mathbf{W}_T^\Delta = \{W^\Delta(t) : 0 \leq t \leq T\}$. We are interested in the following discretization error.

$$\begin{aligned} d(\mathbf{W}_T, \mathbf{W}_T^\Delta) &= \max_{0 \leq t \leq T} |W(t) - W^\Delta(t)| \\ &\leq \max_{0 \leq k \leq \lceil T/\Delta \rceil} \max_{t \in [k\Delta, (k+1)\Delta)} |W(t) - W(k\Delta)|. \end{aligned}$$

Define $V(\Delta) = \max_{0 \leq t \leq \Delta} |W(t)|$. By the Markov property of Brownian motion, we have that

$$d(\mathbf{W}_T, \mathbf{W}_T^\Delta) \leq \max_{1 \leq k \leq \lceil T/\Delta \rceil} V_k(\Delta),$$

where $\{V_k(\Delta) : k \in \{1, \dots, \lceil T/\Delta \rceil\}\}$ is an independent sequence of copies of $V(\Delta)$. From equation (8.3) in Karatzas and Shreve (2012), an upper bound on the tail probabilities of $V(\Delta)$ can be derived. In particular, for $z > 0$,

$$\mathbb{P}(V(\Delta) > z) \leq \frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)}.$$

It follows that, for $z > 0$,

$$\begin{aligned} \mathbb{P}(d(\mathbf{W}_T, \mathbf{W}_T^\Delta) > z) &= 1 - \mathbb{P}(V(\Delta) \leq z)^{T(\Delta)} \leq 1 - \left(1 - \frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)} \right)^{T(\Delta)} \\ &= 1 - \exp \left[T(\Delta) \log \left(1 - \frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)} \right) \right], \end{aligned} \quad (8.6)$$

where $T(\Delta) = \lfloor T/\Delta \rfloor$. Next, we use the inequality that $\log(z) \geq -(1-z)/z$ for $z \in (0, 1)$ to conclude that

$$\mathbb{P}(d(\mathbf{W}_T, \mathbf{W}_T^\Delta) > z) \leq 1 - \exp \left(-T(\Delta) \left(\frac{\frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)}}{1 - \frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)}} \right) \right).$$

Considering the exponent in the preceding equation, we have

$$\begin{aligned} -T(\Delta) \left(\frac{\frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)}}{1 - \frac{4}{z} \sqrt{\frac{\Delta}{2\pi}} e^{-z^2/(2\Delta)}} \right) &= -\left(\frac{T}{\Delta} + O(1) \right) \frac{4\sqrt{\Delta}}{\sqrt{2\pi z e^{z^2/(2\Delta)}} - 4\sqrt{\Delta}} \\ &= -(1 + O(\Delta)) \frac{4T}{\sqrt{2\pi \Delta z e^{z^2/(2\Delta)}} - 4\Delta}, \end{aligned}$$

and thus,

$$\mathbb{P}(d(\mathbf{W}_T, \mathbf{W}_T^\Delta) > z) \leq 1 - \exp \left(-\frac{(1 + O(\Delta))4T}{\sqrt{2\pi \Delta z e^{z^2/(2\Delta)}} - 4\Delta} \right) \leq \frac{(1 + O(\Delta))4T}{\sqrt{2\pi \Delta z e^{z^2/(2\Delta)}} - 4\Delta}. \quad (8.7)$$

Now, consider a scaled version of $\mathbf{W}_T$ defined by $W_n(t) = \frac{1}{n}W(nt)$, $t \in [0, T]$, which is denoted by $\mathbf{W}_{n,T} = \{W_n(t) : 0 \leq t \leq T\}$. For $\Delta > 0$, we consider the discretization $\mathbf{W}_{n,T}^\Delta = \{W_n^\Delta(t) : 0 \leq t \leq T\}$ defined by $W_n^\Delta(t) = W_n((k-1)\Delta)$ for $t \in [(k-1)\Delta, k\Delta)$. Finally, we observe that

$$d(\mathbf{W}_{n,T}, \mathbf{W}_{n,T}^\Delta) = \frac{1}{n} d(\mathbf{W}_{n,T}, \mathbf{W}_{n,T}^{n\Delta}).$$

Therefore, we can use (8.7) to obtain

$$\mathbb{P}(d(\mathbf{W}_{n,T}, \mathbf{W}_{n,T}^\Delta) > z) = \mathbb{P}(d(\mathbf{W}_{n,T}, \mathbf{W}_{n,T}^{n\Delta}) > nz) \leq \frac{(1 + O(n\Delta))4nT}{\sqrt{2\pi n\Delta nze^{nz^2/(2\Delta)}} - 4n\Delta}. \quad (8.8)$$

Under Assumption 1, the Skorokhod map is Lipschitz continuous, and there exists a constant C that only depends on R , such that, for $t \geq 0$,

$$\sup_{0 \leq s \leq t} |Z(s) - Z^\Delta(s)| \leq C \sup_{0 \leq s \leq t} |\tilde{X}(s) - \tilde{X}^\Delta(s)|. \quad (8.9)$$

For $z > 0$, we have

$$\begin{aligned} \mathbb{P} \left(\sup_{0 \leq s \leq t} |Z(s) - Z^\Delta(s)| > z \right) &\leq \mathbb{P} \left(\sup_{0 \leq s \leq t} |\tilde{X}(s) - \tilde{X}^\Delta(s)| > z/C \right) \\ &\leq \mathbb{P} \left(\sup_{0 \leq s \leq t} |X(s) - X^\Delta(s)| > z/C - \Delta|\theta| \right) \\ &\leq \mathbb{P} \left(\sup_{0 \leq s \leq t} |W(s) - W^\Delta(s)| > \frac{z/C - \Delta|\theta|}{d|\Sigma^{1/2}|} \right) \end{aligned}$$

Denote $\tilde{z}_\Delta = (z/C - \Delta|\theta|)/(d|\Sigma^{1/2}|)$. From (8.7), we have

$$\begin{aligned} \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z(s) - Z^\Delta(s)| > z \right) &\leq \mathbb{P} \left(\sup_{0 \leq s \leq T} |W(s) - W^\Delta(s)| > \tilde{z}_\Delta \right) \\ &\leq \frac{(1 + O(\Delta))4T}{\sqrt{2\pi \Delta \tilde{z}_\Delta e^{\tilde{z}_\Delta^2/(2\Delta)}} - 4\Delta}. \end{aligned} \quad (8.10)$$

Now, consider the large derivation scaled processes $Z_n(t) = Z(nt)/n$, $X_n(t) = X(nt)/n$, and $Y_n(t) = Y(nt)/n$. From the homogeneity property of the Skorokhod map for linear scalings, (Z_n, Y_n) is the solution of the Skorokhod problem for $\{\tilde{X}_n(t) = X_n(t) + \theta t; t \geq 0\}$. It follows from (8.8) that

$$\begin{aligned} \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > z \right) &\leq \mathbb{P} \left(\sup_{0 \leq s \leq T} |W_n(s) - W_n^\Delta(s)| > \tilde{z}_\Delta \right) \\ &\leq \frac{(1 + O(n\Delta))4nT}{\sqrt{2\pi n\Delta n\tilde{z}_\Delta e^{n\tilde{z}_\Delta^2/(2\Delta)}} - 4n\Delta}. \end{aligned}$$

Hence,

$$\begin{aligned}
 & \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > z \right) \\
 & \leq \frac{1}{n} \log \left(\frac{(1 + O(n\Delta))4nT}{\sqrt{2\pi n\Delta} n\tilde{z}_\Delta e^{n\tilde{z}_\Delta^2/(2\Delta)} - 4n\Delta} \right) \\
 & = \frac{\log((1 + O(n\Delta))4nT)}{n} - \frac{1}{n} \log \left(\sqrt{2\pi n\Delta} n\tilde{z}_\Delta e^{n\tilde{z}_\Delta^2/(2\Delta)} - 4n\Delta \right) \\
 & = \frac{\log((1 + O(n\Delta))4nT)}{n} - \frac{1}{n} \left[\log(e^{n\tilde{z}_\Delta^2/(2\Delta)}) + \log(n) + \log(\sqrt{2\pi n\Delta}\tilde{z}_\Delta - 4\Delta e^{-n\tilde{z}_\Delta^2/(2\Delta)}) \right] \\
 & = \frac{\log((1 + O(n\Delta))4nT)}{n} - \frac{\tilde{z}_\Delta^2}{2\Delta} - \frac{1}{n} \left[\log(n) + \log(\sqrt{2\pi n\Delta}\tilde{z}_\Delta - 4\Delta e^{-n\tilde{z}_\Delta^2/(2\Delta)}) \right]. \tag{8.11}
 \end{aligned}$$

When $\Delta = \Delta(n) = o(1)$, we have

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > z \right) = -\infty. \quad \square$$

8.5. Proof of Proposition 5

Proposition 5. Let $\eta \in (0, 1/2)$ and $\Delta \equiv \Delta(n)$ satisfy $\lim_{n \rightarrow \infty} n\Delta^\eta = 0$ and assume that $z_n \notin A_n \cup B$ satisfying $\lim_{n \rightarrow \infty} z_n = 0$. Then, under Assumptions 1 and 2,

$$\left| \frac{\max_{x \in C_n} p_n^\Delta(x)}{\max_{x \in C_n} p_n(x)} - 1 \right| = O(n\Delta^\eta), \tag{5.2}$$

where $C_n = \{x \in \mathbb{R}_+^d : \sum_{k=1}^d x_k = \sum_{k=1}^d z_{n,k}\}$ with $z_{n,k}, k = 1, \dots, d$, being the k th component of z_n .

We first introduce the following lemma.

Lemma 7. If $\Delta(n) \rightarrow 0$ as $n \rightarrow \infty$, then for $\eta \in (0, 1/2)$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq \tau_n} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) = -\infty. \tag{8.12}$$

Proof. From (8.10) in the proof of Proposition 4, letting $\xi = \frac{\Delta^\eta/C - \Delta|\theta|}{d|\Sigma^{1/2}|}$, we have that, for $T \geq 0$,

$$\begin{aligned}
 & \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) \\
 & \leq \frac{\log(4nT)}{n} - \frac{\xi^2}{2\Delta} - \frac{1}{n} \left[\log(n) + \log(\sqrt{2\pi n\Delta}\xi - 4\Delta e^{-n\xi^2/(2\Delta)}) \right].
 \end{aligned}$$

Next, noting that

$$\lim_{n \rightarrow \infty} \frac{\xi^2}{2\Delta} = \infty,$$

it follows that

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) = -\infty. \tag{8.13}$$

We next observe that, for $T > 0$,

$$\begin{aligned}
 & \mathbb{P} \left(\sup_{0 \leq s \leq \tau_n} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) \\
 & \leq \mathbb{P} \left(\sup_{0 \leq s \leq T} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) + \mathbb{P}(\tau_n > T).
 \end{aligned}$$

Therefore, by lemma 1.2.16 in Dembo and Zeitouni (2010),

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P} \left(\sup_{0 \leq s \leq \tau_n} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta \right) \\ & \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\tau_n > T). \end{aligned}$$

The result then follows from Proposition 1 and sending $T \rightarrow \infty$. $\square$

We next introduce the following notation. Let $\eta \in (0, 1/2)$ and take $\Delta \equiv \Delta(n)$ small enough such that $\epsilon/n - \Delta^\eta > 0$ and $\epsilon/n + \Delta^\eta < z_n$. Now, define

$$A_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \epsilon/n - \Delta^\eta \right\}, \text{ and } B_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq 1 - \Delta^\eta \right\},$$

and

$$\begin{aligned} \tau_n^-(A_n^-) &= \inf \{ t \geq 0 : Z_n(t) \in A_n^- \}, \quad \tau_n^-(B_n^-) = \inf \{ t \geq 0 : Z_n(t) \in B_n^- \}, \\ \tau_n^- &= \inf \{ t \geq 0 : Z_n(t) \in A_n^- \cup B_n^- \}. \end{aligned}$$

Finally, for $\tilde{z} \in \mathbb{R}_+^d$, let $p_n^-(\tilde{z}) = \mathbb{P}(\tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z})$.

Lemma 8. For $\eta \in (0, 1/2)$, let $\Delta \equiv \Delta(n)$ satisfy $\lim_{n \rightarrow \infty} n\Delta^\eta = 0$. Then,

$$\frac{\max_{\tilde{z} \in \mathcal{C}_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in \mathcal{C}_n} p_n(\tilde{z})} \leq 1 + O(n\Delta^\eta),$$

where $\mathcal{C}_n = \{ \tilde{z} \in \mathbb{R}_+^d : \sum_{i=1}^d \tilde{z}_i = \sum_{i=1}^d z_{n,i} \}$.

Proof. Define the boundaries of A_n^-, B_n^- and A_n inside $\mathbb{R}_+^d$ as follows:

$$\mathcal{A}_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k = \epsilon/n - \Delta^\eta \right\}, \mathcal{B}_n^- = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k = 1 - \Delta^\eta \right\},$$

and

$$\mathcal{A}_n = \left\{ z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k = \epsilon/n \right\}.$$

Using the Markov property,

$$\begin{aligned} & p_n^-(z_n) - p_n(z_n) \\ &= \mathbb{P}(\tau_n^-(B_n^-) < \tau_n^-(A_n^-), \tau_n(B_n^-) > \tau_n(A_n) | Z_n(0) = z_n) \\ &= \mathbb{P}(\tau_n^-(B_n^-) < \tau_n(A_n) < \tau_n(B_n^-) | Z_n(0) = z_n) \\ &\quad + \mathbb{P}(\tau_n(A_n) < \tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = z_n) \\ &\leq \max_{\tilde{z} \in \mathcal{B}_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B_n^-) | Z_n(0) = \tilde{z}) \mathbb{P}(\tau_n^-(B_n^-) < \tau_n(A_n) | Z_n(0) = z_n) \\ &\quad + \max_{\tilde{z} \in \mathcal{A}_n} \mathbb{P}(\tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}) \mathbb{P}(\tau_n(A_n) < \tau_n^-(B_n^-) | Z_n(0) = z_n). \end{aligned}$$

We observe that $\{\tau_n^-(B_n^-) < \tau_n(A_n) | Z_n(0) = z_n\} \subset \{\tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = z_n\}$, and thus,

$$\mathbb{P}(\tau_n^-(B_n^-) < \tau_n(A_n) | Z_n(0) = z_n) \leq p_n^-(z_n).$$

Next, note that, for $\tilde{z} \in \mathcal{A}_n$,

$$\begin{aligned} & \max_{\tilde{z} \in \mathcal{A}_n} \mathbb{P}(\tau_n^-(B_n^-) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}) \\ & \leq \max_{\tilde{z} \in \mathcal{C}_n} p_n^-(\tilde{z}) \max_{\tilde{z} \in \mathcal{A}_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}), \end{aligned}$$

where $\tau_n(C_n) = \inf \{t \geq 0 : Z_n(t) \in C_n\}$. It follows that

$$\begin{aligned} p_n^-(z_n) - p_n(z_n) &\leq p_n^-(z_n) \max_{\tilde{z} \in B_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B) | Z_n(0) = \tilde{z}) \\ &\quad + \max_{\tilde{z} \in C_n} p_n^-(\tilde{z}) \max_{\tilde{z} \in A_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}), \end{aligned}$$

which yields that

$$\begin{aligned} p_n^-(z_n) &\left[1 - \max_{\tilde{z} \in B_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B) | Z_n(0) = \tilde{z}) \right] \\ &\leq p_n(z_n) + \max_{\tilde{z} \in C_n} p_n^-(\tilde{z}) \max_{\tilde{z} \in A_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}). \end{aligned}$$

Taking the maximum over $z_n \in C_n$, we have

$$\begin{aligned} \max_{\tilde{z} \in C_n} p_n^-(\tilde{z}) &\left[1 - \max_{\tilde{z} \in B_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B) | Z_n(0) = \tilde{z}) \right] \\ &\leq \max_{\tilde{z} \in C_n} p_n(\tilde{z}) + \max_{\tilde{z} \in C_n} p_n^-(\tilde{z}) \max_{\tilde{z} \in A_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}), \end{aligned}$$

and

$$\frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \leq \frac{1}{1 - \max_{\tilde{z} \in B_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B) | Z_n(0) = \tilde{z}) - \max_{\tilde{z} \in A_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z})}. \quad (8.14)$$

Similar to the proof of Theorem 1, using the minimality property of the one-dimensional Skorokhod map Γ_1 , we have

$$\begin{aligned} \max_{\tilde{z} \in B_n^-} \mathbb{P}(\tau_n(A_n) < \tau_n(B) | Z_n(0) = \tilde{z}) \\ \leq \mathbb{P}\left(\Gamma_1\left(1 - \Delta^\eta + \sum_{k=1}^d \tilde{X}_k\right) \text{ reaches } \epsilon/n \text{ before reaching } 1\right) \end{aligned} \quad (8.15)$$

$$\begin{aligned} &= \mathbb{P}\left(1 - \Delta^\eta + \sum_{k=1}^d \tilde{X}_k \text{ reaches } \epsilon/n \text{ before reaching } 1\right) \\ &= \frac{1 - e^{-n\gamma\Delta^\eta}}{e^{n\gamma a_n} - e^{-n\gamma\Delta^\eta}}, \end{aligned} \quad (8.16)$$

where $\gamma = \frac{2|\sum \mu_l|}{1 - \sum 1}$ and $a_n = 1 - \Delta^\eta - \epsilon/n$.

We next use the Lyapunov function L introduced in Theorem 3 to derive an upper bound for $\max_{\tilde{z} \in A_n} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z})$. Choose $\kappa_1 > 0$ such that, when $\sum_{i=1}^d z_i \geq \kappa_1$, $\text{tr}(\ddot{L}(z)\Sigma) < c_0$. Let $\kappa_2 = 2\kappa_1/\epsilon$ and define $Z(t) = \kappa_2 Z(t)$ for $t \geq 0$. Then, for $\tilde{z} \in A_n$,

$$\begin{aligned} \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}) \\ = \mathbb{P}(\check{Z}(\cdot) \text{ first reaches } \kappa_2 n C_n \text{ before reaching } \kappa_2 n A_n^- | \check{Z}(0) = \kappa_2 n \tilde{z}). \end{aligned}$$

From Theorem 3(v), for $\check{z} \in \mathbb{R}_+^d$, $L(\check{z}) = (\sum_{i=1}^d \check{z}_i) L(\check{z}/\sum_{i=1}^d \check{z}_i)$. Let $\underline{l} = \inf \{L(z) : z \in \mathbb{R}_+^d, \sum_{i=1}^d z_i = 1\}$. Then, $\underline{l} \in (0, \infty)$, and

$$\sum_{i=1}^d \check{z}_i \leq \frac{L(\check{z})}{\underline{l}}. \quad (8.17)$$

We further define the following two sets

$$\begin{aligned} \tilde{A} &= \left\{ \check{z} \in \mathbb{R}_+^d : L(\check{z}) \leq \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \right\}, \\ \tilde{C} &= \left\{ \check{z} \in \mathbb{R}_+^d : L(\check{z}) \geq \underline{l} \kappa_2 \sum_{i=1}^d z_i \right\}. \end{aligned}$$

We see that $\tilde{A} \subset \kappa_2 n A_n^-$ and $\kappa_2 n C_n \subset \tilde{C}$, where $C_n = \{\check{z} \in \mathbb{R}_+^d : \sum_{i=1}^d \check{z}_i \geq \sum_{i=1}^d z_{n,i}\}$. Indeed, for $\check{z} \in \tilde{A}$, from (8.17),

$$\sum_{i=1}^d \check{z}_i \leq \frac{L(\check{z})}{\underline{l}} \leq \underline{l} \frac{\kappa_2 (\epsilon - n \Delta^\eta)}{\underline{l}} = \kappa_2 n (\epsilon/n - \Delta^\eta),$$

which says $\check{z} \in \kappa_2 n A_n^-$. Now, for $\check{z} \in \kappa_2 n C_n$, that is $\sum_{i=1}^d \check{z}_i \geq \kappa_2 \sum_{i=1}^d z_i$, again from (8.17), we have

$$L(\check{z}) \geq \underline{l} \sum_{i=1}^d \check{z}_i \geq \underline{l} \kappa_2 \sum_{i=1}^d z_i.$$

Fix $\tilde{z} \in A_n$. Hence, from (8.17),

$$\begin{aligned} & \mathbb{P}(\tau_n(C_n) < \tau_n^-(A_n^-) | Z_n(0) = \tilde{z}) \\ & \leq \mathbb{P}(\check{Z}(\cdot) \text{ first reaches } \tilde{C} \text{ before reaching } \tilde{A} | \check{Z}(0) = \kappa_2 n \tilde{z}) \\ & \leq \mathbb{P}(L(\check{Z}(\cdot)) \text{ first reaches } \underline{l} \kappa_2 \sum_{i=1}^d z_i \text{ before reaching } \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) | \check{Z}(0) = \kappa_2 n \tilde{z}) \\ & \equiv \mathbf{p}(\tilde{z}). \end{aligned}$$

We next consider the following martingale: for $\lambda > 0$,

$$U(t) = \exp \left\{ \lambda L(\check{Z}(t)) - \lambda c(t) - \frac{\lambda^2}{2} \int_0^t |\dot{L}(\check{Z}(s))|^2 ds \right\},$$

where

$$c(t) = \int_0^t \frac{1}{2} \text{tr}(\ddot{L}(\check{Z}(s))) ds + \int_0^t \sum_{i=1}^d \dot{L}_i(\check{Z}(s)) \theta_i ds + \sum_{i=1}^d \int_0^t \sum_{j=1}^d \dot{L}_j(\check{Z}(s)) R_{ij} dY_i(s).$$

Let $\tau(\tilde{z}) = \inf \{t \geq 0 : \kappa_3 L(\check{Z}(t)) \in \tilde{A} \cup \tilde{C}\}$. Now let $m = n = 1$ and $\Delta = t$ in (8.1) in the proof of Proposition 1, and we have for positive constants κ_4 ,

$$c(t) \leq -\kappa_4 t, \quad 0 \leq t \leq \tau(\tilde{z}).$$

From Theorem 3(vii), $|\dot{L}(Z(t))| \leq M_4$, and we can choose λ such that

$$\lambda c(t) + \frac{\lambda^2}{2} \int_0^t |\dot{L}(\check{Z}(s))|^2 ds \leq 0, \quad 0 \leq t \leq \tau(\tilde{z}).$$

Using the optional sampling theorem, we have

$$\begin{aligned} \exp \{ \lambda L(\kappa_2 n \tilde{z}) \} &= \mathbb{E}[U(0)] = \mathbb{E}[U(\tau(\tilde{z}))] \\ &= \mathbf{p}(\tilde{z}) \mathbb{E} \left[U(\tau(\tilde{z})) | L(\check{Z}(\tau(\tilde{z}))) = \underline{l} \kappa_2 \sum_{i=1}^d z_i \right] \\ &\quad + (1 - \mathbf{p}(\tilde{z})) \mathbb{E}[U(\tau(\tilde{z})) | L(\check{Z}(\tau(\tilde{z}))) = \underline{l} \kappa_2 (\epsilon - n \Delta^\eta)] \\ &\geq \mathbf{p}(\tilde{z}) \exp \left\{ \lambda \underline{l} \kappa_2 \sum_{i=1}^d z_i \right\} + (1 - \mathbf{p}(\tilde{z})) \exp \{ \lambda \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \}. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbf{p}(\tilde{z}) &\leq \frac{\exp \{ \lambda L(\kappa_2 n \tilde{z}) \} - \exp \{ \lambda \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \}}{\exp \{ \lambda \underline{l} \kappa_2 \sum_{i=1}^d z_i \} - \exp \{ \lambda \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \}} \\ &\leq \frac{\exp \{ \lambda \kappa_2 \epsilon L(n \tilde{z} / \epsilon) \} - \exp \{ \lambda L(n \tilde{z} / \epsilon) \kappa_2 (\epsilon - n \Delta^\eta) \}}{\exp \{ \lambda \underline{l} \kappa_2 \sum_{i=1}^d z_i \} - \exp \{ \lambda \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \}} \\ &\leq (1 - \exp \{ -n \Delta^\eta \lambda \underline{l} \kappa_2 \}) \frac{\exp \{ \lambda \kappa_2 \epsilon L(n \tilde{z} / \epsilon) \}}{\exp \{ \lambda \underline{l} \kappa_2 \sum_{i=1}^d z_i \} - \exp \{ \lambda \underline{l} \kappa_2 (\epsilon - n \Delta^\eta) \}} \\ &= O(n \Delta^\eta). \end{aligned} \tag{8.18}$$

Applying (8.18) and (8.16) to (8.14), we have that

$$\frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \leq 1 + O(n\Delta^\eta).$$

This completes the proof. $\square$

Proof of Proposition 5. We note that

$$\begin{aligned} & \frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1 \\ &= \frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})} \frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} - 1. \end{aligned}$$

Recall that $\tau_n^- = \inf\{t \geq 0 : \sum_{k=1}^d Z_{n,k}(t) \geq 1 - \Delta^\eta, \text{ or } \sum_{k=1}^d Z_{n,k}(t) \leq \epsilon/n - \Delta^\eta\}$, and $p_n^-(z_n) = \mathbb{P}(\sum_{k=1}^d Z_{n,k}(\tau_n^-) = 1 - \Delta^\eta | Z_n(0) = z_n)$. We note that Lemma 7 is still valid when τ_n is replaced by τ_n^- . Thus, from the proof of Lemma 7, we have

$$\mathbb{P}\left(\sup_{0 \leq s \leq \tau_n^-} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right) = e^{-n\xi^2/(2\Delta) + o(n)},$$

where $\xi = \frac{\Delta^\eta/C - \Delta|\theta|}{d|\Sigma^{1/2}|}$. Also Theorem 1 is valid for $p_n^-(z_n)$, which yields that

$$p_n^-(z_n) = e^{-n \inf_{x \in B} I(x) + o(n)}. \quad (8.19)$$

We observe that

$$\begin{aligned} p_n^\Delta(z_n) &= \mathbb{P}\left(\tau_n^\Delta(B_n) < \tau_n^\Delta(A_n), \sup_{0 \leq s \leq \tau_n^-} |Z_n(s) - Z_n^\Delta(s)| < \Delta^\eta\right) \\ &\quad + \mathbb{P}\left(\tau_n^\Delta(B_n) < \tau_n^\Delta(A_n), \sup_{0 \leq s \leq \tau_n^-} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right) \\ &\leq p_n^-(z_n) + \mathbb{P}\left(\sup_{0 \leq s \leq \tau_n^-} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right) \end{aligned}$$

Noting that $\frac{\xi^2}{2\Delta} = \Theta(\Delta^{2\eta-1})$, for some constant $C > 0$, we have

$$\frac{p_n^\Delta(z_n)}{p_n^-(z_n)} \leq 1 + \frac{\mathbb{P}\left(\sup_{0 \leq s \leq \tau_n^-} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right)}{p_n^-(z_n)} = 1 + o(e^{-Cn\Delta^{2\eta-1}}). \quad (8.20)$$

Finally, from Lemma 8, we obtain

$$\frac{\max_{\tilde{z} \in C_n} p_n^-(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \leq 1 + O(n\Delta^\eta).$$

Combining these estimates, we establish that

$$\frac{\max_{\tilde{z} \in C_n} p_n^\Delta(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \leq 1 + O(n\Delta^\eta).$$

To show the other direction, we define $A_n^+ = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \leq \epsilon/n + \Delta^\eta\}$, $B_n^+ = \{z \in \mathbb{R}_+^d : \sum_{k=1}^d z_k \geq 1 + \Delta^\eta\}$, $\tau_n^+ = \inf\{t \geq 0 : Z_n(t) \in A_n^+ \cup B_n^+\}$, $\tau_n^+(B_n^+) = \inf\{t \geq 0 : Z_n(t) \in B_n^+\}$, $\tau_n^+(A_n^+) = \inf\{t \geq 0 : Z_n(t) \in A_n^+\}$, and $p_n^+(z_n) = \mathbb{P}(\tau^+(B_n^+) < \tau^+(A_n^+) | Z_n(0) = z_n)$. Similar to Lemma 8 (replace $p_n^-(z_n)$ and $p_n(z_n)$ by $p_n(z_n)$ and $p_n^+(z_n)$, respectively, in the proof), it can be shown that

$$\frac{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \geq 1 + O(n\Delta^\eta).$$

Further note that

$$\begin{aligned}
 p_n^\Delta(z_n) &= \mathbb{P}\left(\tau_n^\Delta(B_n) < \tau_n^\Delta(A_n), \sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| < \Delta^\eta\right) \\
 &\quad + \mathbb{P}\left(\tau_n^\Delta(B_n) < \tau_n^\Delta(A_n), \sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right) \\
 &\geq \mathbb{P}\left(\tau_n^\Delta(B_n) < \tau_n^\Delta(A_n), \sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| < \Delta^\eta\right) \\
 &\geq \mathbb{P}\left(\tau_n^+(B_n^+) < \tau_n^+(A_n^+), \sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| < \Delta^\eta\right) \\
 &\geq p_n^+(z_n) - \mathbb{P}\left(\sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right).
 \end{aligned}$$

Similar to (8.19) and (8.20), we have, for each $\tilde{z} \in C_n$,

$$p_n^+(\tilde{z}) = e^{-n \inf_{x \in B} I(x) + o(n)},$$

and

$$\frac{p_n^\Delta(z_n)}{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})} \geq 1 - \frac{\mathbb{P}\left(\sup_{0 \leq s \leq \tau_n^+} |Z_n(s) - Z_n^\Delta(s)| > \Delta^\eta\right)}{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})} = 1 - o(e^{-Cn\Delta^{2\eta-1}}).$$

Finally, we have

$$\frac{\max_{\tilde{z} \in C_n} p_n^\Delta(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} = \frac{\max_{\tilde{z} \in C_n} p_n^\Delta(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})} \frac{\max_{\tilde{z} \in C_n} p_n^+(\tilde{z})}{\max_{\tilde{z} \in C_n} p_n(\tilde{z})} \geq 1 + O(n\Delta^\eta). \quad \square$$

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Appendix

All other proofs are collected in Section A.1. In Section A.2, we construct the explicit solution of the VP for two-dimensional SRBMs following Avram et al. (2001).

A.1. Additional Proofs

Proposition 2. Given $K \subseteq L$ and two points $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1 with respect to K , there exists a unique $J_* \subseteq K$ such that $\langle A^{J_*} R_j, \alpha^{J_*}(w, v)(v - w) - \theta \rangle_{\Sigma^{-1}} \leq 0$, $j \in K \setminus J_*$, and when $J_* \neq \emptyset$, each component of $\lambda^{J_*}(w, v)$ is positive. Furthermore,

$$\begin{aligned}
 \mathcal{I}_K^*(w, v) &= \langle b_K(w, v) - \theta, v - w \rangle_{\Sigma^{-1}} \\
 &= |A^{J_*} \theta|_{\Sigma^{-1}} |A^{J_*}(v - w)|_{\Sigma^{-1}} - \langle A^{J_*} \theta, A^{J_*}(v - w) \rangle_{\Sigma^{-1}},
 \end{aligned}$$

and an optimal path is

$$x_K(t) = w + tb_K(w, v), \quad 0 \leq t \leq \alpha^{J_*}(w, v),$$

where

$$b_K(w, v) = \begin{cases} \alpha^{J_*}(w, v)(v - w) - R_{LJ_*} \alpha^{J_*}(w, v) B^{J_*}(v - w) + R_{LJ_*} B^{J_*} \theta, & \text{when } J_* \neq \emptyset, \\ \alpha^{J_*}(w, v)(v - w), & \text{when } J_* = \emptyset. \end{cases}$$

Proof. Proposition 2 summarizes some major results of El Kharroubi et al. (2012), including theorems 3 and 4, lemma 5, and propositions 1 and 2. Despite a small difference in problem settings, their proofs are still valid for our setting. To be more specific, given $K \subseteq L$ and two points $v, w \in \mathbb{R}_+^d$ that satisfy Condition 1 with respect to K , theorems 3 and 4 of El Kharroubi et al. (2012) assert that there is a unique $J_* \subseteq K$ such that $\langle A^{J_*} R_j, \alpha^{J_*}(w, v)(v - w) - \theta \rangle_D \leq 0$, $j \in K \setminus J_*$, and when

$J_* \neq \emptyset$, each component of $\lambda^J(w, v)$ is positive. Proposition 1 of El Kharroubi et al. (2012) shows that $\mathcal{I}_K^*(w, v) = |A^J \theta|_D |A^J(v - w)|_D - \langle A^J \theta, A^J(v - w) \rangle_D$. Lemma 5 and proposition 2 of El Kharroubi et al. (2012) give the explicit form of an optimal path. $\square$

Proposition 6. *The splitting algorithm for Z_n^Δ is stable; that is, the total number of particles ever used grows subexponentially as $n \rightarrow \infty$. Letting $\{s_n^{(k)}\}_{k \in \mathbb{N}}$ be an independent and identically distributed sequence of copies of s_n , then for each $N \in \mathbb{N}$,*

$$\mathbb{E} \left[\frac{1}{N} \sum_{k=1}^N s_n^{(k)} \right] = p_n^\Delta(z_n). \quad (5.7)$$

Furthermore, the variance of s_n can be measured using the following rate of decay of the second moment:

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}_{z_n} [s_n^2] = \inf_{x \in B} I(x) - \bar{T}(0). \quad (5.8)$$

Proof. The stability follows directly from proposition 7 in Dean and Dupuis (2009), and (5.7) follows from lemma 5 in the same reference. To prove (5.8), we need to verify that conditions 1 and 4 in Dean and Dupuis (2009) hold for our problem. Condition 1 of Dean and Dupuis (2009) follows from Theorem 1. For their condition 4, because of the positive recurrence of our defined process, σ_n is finite almost surely. Then, by strong Markov property,

$$1_{\{Z_n(\sigma_n) \in L_x\}} (p_n(Z_n(\sigma_n)))^2 \leq 1_{\{Z_n(\sigma_n) \in L_x\}} (p_n(Z_n(\sigma_n))) \sup_{y \in \partial L_x} p^n(y).$$

Hence, using the strong Markov property of Z_n , we have

$$\begin{aligned} \mathbb{E}_{z_n} [1_{\{Z_n(\sigma_n) \in L_x\}} (p_n(Z_n(\sigma_n)))^2] &\leq \sup_{y \in \partial L_x} p_n(y) \cdot \mathbb{E}_{z_n} [1_{\{Z_n(\sigma_n) \in L_x\}} p_n(Z_n(\sigma_n))] \\ &= \sup_{y \in \partial L_x} p^n(y) \cdot p_n(z_n). \end{aligned}$$

By the large deviation upper bound, we can get

$$\begin{aligned} &\liminf_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}_{z_n} [1_{\{Z_n(\sigma_n) \in L_x\}} (p_n(Z_n(\sigma_n)))^2] \\ &= -\limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_{z_n} [1_{\{Z_n(\sigma_n) \in L_x\}} (p_n(Z_n(\sigma_n)))^2] \\ &\geq -\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left[\sup_{y \in \partial L_x} p_n(y) \cdot p_n(z_n) \right] \\ &= -\limsup_{n \rightarrow \infty} \frac{1}{n} \left[\log \sup_{y \in \partial L_x} p_n(y) + \log p_n(z_n) \right] \\ &\geq -\limsup_{n \rightarrow \infty} \frac{1}{n} \log \sup_{y \in \partial L_x} p_n(y) - \limsup_{n \rightarrow \infty} \frac{1}{n} \log p_n(z_n) \\ &\geq \inf_{y \in \partial L_x} W(y) + W(z). \end{aligned}$$

Now, (5.8) follows directly from theorem 8 in Dean and Dupuis (2009). $\square$

Proposition 7. *The following hold for the RESTART algorithm:*

- i. *The estimator constructed from the RESTART algorithm is unbiased.*
- ii. *Suppose that $z_n \rightarrow 0$ with each $z_n \notin (A_n \cup B)$. Then,*

$$\liminf_{n \rightarrow \infty} -\frac{1}{n} \log \mathbb{E}_{z_n} [r_n^2] \geq \inf_{y \in B} \{I(y) + (\bar{T}(0) - \bar{T}(y)) \vee 0\}.$$

- iii. *Let w^n denote the sum of the lifetimes of all the particles simulated in a single trial. Then,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_{z_n} [w^n] = 0.$$

Proof. It suffices to verify that the five conditions in Dean and Dupuis (2011), which are conditions 3.1, 4.3, 5.1, 5.2, and 5.3, are satisfied. We note that, for our specific problem, those conditions essentially require that

- A_n and B are closed sets.
- $Z_n^\Delta(t)$ satisfies the large deviation principle.
- For any compact $K \subset \mathbb{R}_+^d$, all zero cost trajectories with initial positions in K enter $(A_n \cup B)^\circ$ by some fixed finite time depending on K .

Indeed, all these conditions are satisfied, and thus, the proposition follows. $\square$

A.2. Explicit Solution for the VP in $\mathbb{R}_+^2$

The VP in $\mathbb{R}_+^2$ is analytically solved in Avram et al. (2001), in which the solution is characterized by defining “cones of boundary influence.” In this section, we summarize the result in Avram et al. (2001) and adopt its notation. Without loss of generality, we let

$$R = \begin{bmatrix} 1 & r_2 \\ r_1 & 1 \end{bmatrix}. \quad (\text{A.1})$$

For $i = 1, 2$, we choose a vector p^i that is orthogonal (under the usual Euclidean inner product) to the i th column of R , and $|\Sigma p^i|_{\Sigma^{-1}} = 1$. We recall the *face* $F_i = \{x \in \mathbb{R}_+^2 : x_i = 0\}$, and define the associated *exit velocity* $a^i = \theta - 2(\theta' p^i) \Sigma p^i$. The face F_i is said to be *reflective* if the i th component of a^i is negative, that is, $a_i^i < 0$. Denote $e^1 = (0, 1)^T$ and $e^2 = (1, 0)^T$. Let n^i be a vector that is normal to F_i , pointing to the interior of the state space and normalized such that $|\Sigma n^i|_{\Sigma^{-1}} = 1$. We introduce the key definition from Avram et al. (2001) as follows:

Definition 7 (Cone of Influence).

- i. When F_i is not reflective, the cone of influence C_i is defined to be an empty set. In this case, the face F_i has no boundary influence on solutions to the VP for any $z \in \mathbb{R}_+^2$.
- ii. When F_i is reflective, the cone of influence C_i is defined to be the cone generated by e^i and $\tilde{a}^i$, where

$$\tilde{a}^i = \langle a^i, e^i \rangle_{\Sigma^{-1}} e^i - \langle a^i, \Sigma n^i \rangle_{\Sigma^{-1}} \Sigma n^i.$$

That is,

$$C_i = \{\alpha e^i + \beta \tilde{a}^i : \alpha \geq 0, \beta \geq 0\}.$$

For $z \in \mathbb{R}_+^2$, the cone of influence C_i is said to be *active* if $z \in C_i$. We define three cost functions:

$$I^0(z) = \langle \tilde{a}^0(z) - \theta, z \rangle_{\Sigma^{-1}},$$

$$I^1(z) = \langle \tilde{a}^1 - \theta, z \rangle_{\Sigma^{-1}},$$

$$I^2(z) = \langle \tilde{a}^2 - \theta, z \rangle_{\Sigma^{-1}},$$

where $\tilde{a}^0(z) = z|\theta|_{\Sigma^{-1}}/|z|_{\Sigma^{-1}}$. With Condition (4.1), Avram et al. (2001) shows that an optimal path can always be found among these three paths. The search for such a path depends on the location of z and its active cone of influence. The following is the main result from Avram et al. (2001).

Proposition 8 (Theorem 3.1 of Avram et al. 2001). *Suppose R is completely S and (θ, R) further satisfies Assumption 2. Then,*

- i. *If $z \notin C_1 \cup C_2$, the optimal value of the VP is given by $I^0(z)$.*
- ii. *If $z \in C_1 \setminus C_2$, the optimal value of the VP is given by $I^1(z)$.*
- iii. *If $z \in C_2 \setminus C_1$, the optimal value of the VP is given by $I^2(z)$.*
- iv. *If $z \in C_1 \cap C_2$, the optimal value of the VP is given by $\min\{I^1(z), I^2(z)\}$.*

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